Explainable Ensemble Models for Spatio-Temporal Dengue Outbreak Forecasting in Bangladesh

Submitted by

Ashikur Rahman Ashik
2002026

Sadman Sakib
2002070

A THESIS IN THE FACULTY OF COMPUTER SCIENCE AND ENGINEERING PRESENTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND ENGINEERING

Faculty of Computer Science and Engineering
Patuakhali Science and Technology University
Patuakhali, Bangladesh

February, 2026

DECLARATION

This is to certify that the work presented in this thesis, titled, “Explainable Ensemble Models for Spatio-Temporal Dengue Outbreak Forecasting in Bangladesh”, is the outcome of the investigation and research carried out by us under the supervision of Dr. Md. Abdul Masud, Professor, Department of Computer Science and Information Technology, Patuakhali Science & Technology University.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma or other qualifications.

_____

Ashikur Rahman Ashik
2002026

_____

Md.Sadman Sakib
2002070

CERTIFICATION

This thesis titled, "**Explainable Ensemble Models for Spatio-Temporal Dengue Outbreak Forecasting in Bangladesh**", submitted by the group as mentioned below has been accepted as satisfactory in partial fulfillment of the requirements for the degree B.Sc. in Computer Science and Engineering in February, 2026.

"Board of Examiners"

Supervisor
Dr. Md. Abdul Masud
Professor
Dept. of CSIT

Chairman
Md. Atikqur Rahman
Professor
Dept. of CSIT

Internal Member
Md. Mahbubur Rahman
Associate Professor
Dept. of CSIT

Internal Member
Chinmay Bepery
Professor
Dept. of CSIT

External Member
Dr. Md. Samsuzzaman
Professor
Dept. of CCE

ACKNOWLEDGEMENTS

First and foremost, we express our heartfelt gratitude to Almighty Allah. Without His blessings, completing this research would not have been possible. We would like to extend our sincere thanks to our supervisor, Professor Dr. Md. Abdul Masud, Department of Computer Science and Information Technology, whose unwavering support and invaluable guidance have helped us complete this thesis on time. His insights and advice consistently steered us in the right direction throughout the research process. We are also deeply thankful to our parents, for their unwavering belief in us and for their constant support during the last one years. Their encouragement helped us persevere through challenging times.

Lastly, we wish to thank all the faculty members, staff, and our friends who have provided motivation, mental support, and strength, helping us stay focused and determined to finish this thesis.

Patuakhali
February, 2026

Ashikur Rahman Ashik

Md. Sadman Sakib

CONTENTS

DECLARATION i
CERTIFICATION ii
ACKNOWLEDGEMENTS iii
CONTENTS iv
LIST OF FIGURES vii
LIST OF TABLES viii
ABSTRACT 1
Chapter 1 2
INTRODUCTION 2
1.1 Background of the Study 2
1.2 Motivation 3
1.3 Problem statement 4
1.4 Research Objectives 4
1.5 Thesis Contribution 5
1.6 Thesis Outline 6
Chapter 2 7
LITERATURE REVIEW 7
2.1 Traditional Epidemiological Models for Dengue 7
2.2 Machine Learning in Infectious Disease Forecasting 8
2.3 T-he Role of Meteorological Factors in Dengue Transmission 10
2.4 Explainable AI (XAI) in Healthcare and Epidemiology 10
2.5 Research Gaps in the Context of Bangladesh 12
Chapter 3 14
METHODOLOGY 14
3.1 Introduction 14
3.2 Data Sources 16
3.2.1 DGHS Daily Dengue Patient Records 16
3.2.2 Meteorological Data from Script-Based Retrieval (Open-Meteo and ERA5) 16
3.3 Dataset Characteristics and Exploratory Data Analysis 17

3.3.1 Dataset Overview 17
3.3.2 Temporal Distribution and Zero Inflation 18
3.3.3 Spatial Variability Across Administrative Divisions 20
3.4 Data Cleaning and Pre-processing 20
3.4.1 Handling Missing Values and Data Consistency 21
3.4.2 Target Variable Transformation 21
3.4.3 Temporal Integrity and Leakage Control 22
3.4.4 Feature Scaling 22
3.5 Advanced Feature Engineering 22
3.5.1 Target Variable Transformation (Log-Scaling) 23
3.6 Temporal Feature Extraction 23
3.6.1 Autoregressive Lags (Patients_lag) 24
3.6.2 Rolling Window Statistics 24
3.7 Seasonal Encoding (Fourier Series and Harmonic Regression) 24
3.8 Climate Indices (Heat Index and Rain–Temp Index) 25
3.9 Experimental Setup and Time-Series Train-Test Split (2022–2025) 27
3.10 Modeling and Optimization 29
3.10.1 Selection of Machine Learning Models 29
3.10.2 Linear Baseline: Ridge Regression 29
3.10.3 Ensemble Boosting: LightGBM, CatBoost, and XGBoost 30
3.10.4 Bagging and Kernel Models: Random Forest and SVR 30
3.11 Hyperparameter Optimization (HPO) Strategy 30
3.11.1 Log-Domain Optimization for RMSLE 31
3.11.2 Model-Specific Search Space Design 31
3.11.3 Integrated Constraints and Early Stopping 31
3.11.4 Final Refitting and Efficiency Analysis 32
3.12 Model Evaluation Metrics 32
Chapter 4 36
RESULTS AND PERFORMANCE ANALYSIS 36
4.1 Comparison of Baseline Model Performance 36
4.2 Impact of Hyperparameter Tuning on Model Performance 37
4.3 Analysis of "Accuracy Champion" (LightGBM) 38
4.4 Spatio-Temporal Variability in Model Performance 39
Chapter 5 41
EXPLAINABILITY AND DISCUSSION 41

5.1 Feature Importance Analysis Using SHAP Values 41
5.2 Temporal Fit and Error Analysis 42
5.3 Seasonal and Regional Error Variability 43
5.4 Climatic Drivers of Dengue in Bangladesh 44
5.5 Spatial Burden of Dengue Cases 46
5.6 Limitations of the Study 47
Chapter 6 48
CONCLUSION AND FUTURE WORK 48
6.1 Summary of Findings 48
6.2 Practical Implications for Public Health Authorities 49
6.3 Future Research Directions 50
BIBLIOGRAPHY 54
APPENDICES 57
7.1 Ai & Similarity Test 57
7.2 Weather Data Collection and Processing 58
7.3 Abbreviations and Acronyms 59
7.4 Glossary 60

LIST OF FIGURES

3.1 Overall Methodology of Our Work ……. 16

3.2 Distribution of daily dengue cases showing raw and log-scaled views … 20

3.3 Correlation Heatmap …… 29

4.1 Impact on performance after tuning ……. 39

4.2 SpatioTemporal_Prediction ……... 40

5.1 SHAP Beeswarm Plot …… 42

5.2 5.2a (Observed vs Predicted Cases) & 5.2b (Absolute Error Trends) ……. 43

5.3 Division-wise MAE by Season …….. 44

5.4 Stacked Climate Trends …… 46

5.5 Choropleth Map, (2022–2025) ……... 47

7.1 AI Checking Result using Turnitin ……... 58

7.2 Similarity report of the thesis ……... 59

LIST OF TABLES

3.1 Summary of Data Sources Used in the Study …… 18

3.2 Overview of the Integrated Dengue–Meteorological Dataset ……. 19

3.3 Division-wise Summary Statistics of Dengue Incidence (2022– 2025) … 21

3.4 Feature Engineering for Dengue Outbreak Forecasting …… 27

3.5 Significance of Evaluation Metrics Used in Dengue Forecasting …… 34

4.1 Baseline Model Performance Comparison …….. 36

4.2 Tuned Models Performance Comparison …… 37

6.1 Comparative analysis of state-of-the-art research work in predicting dengue incidents globally using different data sources …….. 52

ABSTRACT

Dengue fever is a significant public health issue in Bangladesh with high seasonality, spatial variability across the eight administrative divisions, and non-linear association of climatic and epidemiologic/case cycles. Classical compartmental and statistical models are frequently inappropriate for high-dimensional surveillance data; therefore, this paper proposes a robust, interpretable machine learning method developed for accurate spatiotemporal prediction to facilitate early warning systems and public health decision making. The study combines daily dengue cases reported by the Directorate General of Health Services (DGHS) with high-resolution meteorological time series from Open-Meteo and ERA5. A dedicated pre-processing pipeline handles missing values, zero inflation and time shifts. Feature engineering consists of autoregressive lags, rolling-window statistics and Fourier-based seasonal representation and domain-informed observables such as the Heat Index and Rain–Temperature Index capturing environmental drivers.For the estimation of incidence dynamics, we performed a systematic benchmark on ensemble/baseline methods: including Ridge Regression, Random Forest, Support Vector Regression and Gradient Boosting Machines (LightGBM, CatBoost and XGBoost).Causality was preserved by applying a time-based train–test split, and model fitness was assessed with MAE, RMSE, RMSLE, and R^2. Experiments demonstrate that boosting ensembles namely CatBoost and LightGBM achieve superior out-of-the-box performance compared to baselines exhibiting best generalization across all splits. SHapley Additive explanations (SHAP) were used to reconcile "black-box" modeling with clinical usefulness. The lagged incidence, rainfall intensity and temperature were found as the main predictors. This investigation offers a science-driven decision support algorithm and reproducible structure for evidence-based intervention in Bangladesh and other dengue endemic areas across the worldwide.

Chapter 1

INTRODUCTION

1.1 Background of the Study

Dengue fever is one of the fastest growing mosquito-borne virus around the globe and has become an endemic in tropical and subtropical regions [1]. For the past 20 years, dengue in Bangladesh has shifted from being an occasional urban health event to a national epidemiological crisis with all eight administrative districts experiencing recent outbreaks [1],[5]. This spatial expansion over wide geographic spread underscores the need for predictive methods that simultaneously take into account temporal dynamics and regional variation.

The dynamics of dengue virus transmission is largely affected by the mosquito vectors, namely Aedes spp., while adult mosquitoes' breeding behaviour and their level competency for transmitting the pathogen is related to climatic factors such as temperature and rainfall [9]. Monsoon-based rainfall, escalating temperature and discontinuous urbanization together provide the favorable conditions for continuous mosquito proliferation which increases outbreak risk in both peri-urban and urban areas of Bangladesh [1]. These interactions between climate and disease are typically non-linear and time-lagged, which greatly complicate predictions.

Historically, dengue prediction models in Bangladesh have been predominantly statistical regression techniques or compartmental epidemiological models [18]. Although these models allow for important theoretical developments, they are often based on linear relationships and uniform transmission patterns characterizing different divisions [16], thus their limited capability to characterize the delayed impact of climatic factors and spatial differences among divisions. Therefore, their prediction performance deteriorates when they are adapted to long-term and high-resolution surveillance data.

Recent innovation in machine learning allows for flexible and data-driven models of complex non-linear relations without stringent a priori assumptions. Ensemble-based models have been shown to work well in dealing with high-dimensional inputs, heterogenous feature

distribution, and temporal dependencies often present in epidemiological data [10]. These features render ensemble learning a favourable framework for spatio-temporal dengue outbreak prediction in data-rich settings, like Bangladesh.

Algorithmic models,however, come with limited interpretability, leading to concerns over its suitability for public health decision-making. Prediction systems for disease surveillance should not only present accurate predictions, but also explain the decisions in a transparent manner consistent with epidemiological knowledge – which has led to development of explainable artificial intelligence (XAI) to be integrated into predictive modeling pipelines [20].

1.2 Motivation

The recent rise in the number and range of dengue outbreaks in Bangladesh has highlighted significant limitations in current surveillance systems and predictive capacity. Even though there is bounty of the epidemiological and meteorological data, only a few operational systems that warn early region-wise reliably are in existence [2],[23]. A number of existing approaches are not regionally or temporally transferable, making them less useful for national-level public health planning.

A major motivation in this work is to model dengue dynamics in a way that explicitly includes the dependence between time and space over eight administrative divisions of Bangladesh. Variation in population density, existence of urban infrastructure, level and manifest climate exposure as well as past disease burden leads to differences in the pattern of outbreaks among regions highlighting the need for spatially aware forecasting systems [14].

Another factor that motivated it is the inadequacy of traditional statistical and epidemiological models for complex dengue data in a real scenario. These models have difficulties including time-lagged climate effects, non-linear interactions, and high-dimensional feature spaces at the same time. Ensemble-based machine learning frameworks (e.g. gradient boosting machines) provide a complementary approach by encoding such relationships implicitly from the data and can enhance the prediction accuracy in spatio-temporal forecasting problems [33].

Its also, the need for interpretability in public health prediction. Prediction without transparent explanation of black-box model undermine trust and adoption by public health

officials [21],[22]. By including explainable modeling methods, predictions can be verified by domain knowledge and leads to evidence-based decisions.

1.3 Problem statement

Even though dengue surveillance and meteorological data is available in Bangladesh, there remains a lack of robust interpretable monitoring systems for spatio-temporal outbreak forecast. Current models often do not deliver consistent division-level predictions over time, which reduces their potential to serve as early warning instruments for public health planners.

The dynamics of dengue transmission are complex, non-linear and delayed, which represents a major difficulty for the conventional modeling. Moreover, inadequate validation methodologies, including random train–test splits, may also produce inflated results that are not consistent with realistic forecasting tasks. In addition, a large proportion of machine learning–derived models focus more on predictive performance rather than interpretability which obstructs their implementation in public health practice.

As a result, the main problem addressed herein is our unavailability of a robust explainable spatio-temporal dengue outbreak forecast system with the ability to accurately predict division-wise dynamics of dengue while respecting temporal causality and being able to produce interpretable insights for public health decision-making [1].

1.4 Research Objectives

The key aim of the proposed project is to build a reliable, interpretable, and space-time explicit forecasting framework of dengue outbreaks for Bangladesh epidemiological and climatic context. The specific objectives are set forth to realize the main objective:

To develop an extensive spatial and temporal dengue dataset for Bangladesh by combining the division-wise daily dengue incidence time series with high-resolution meteorological data, quality control, and ensuring consistency in terms of coverage and temporal alignment of data across all eight administrative divisions.

To compare advanced feature engineering approaches (e.g. autoregressive lag features, rolling-window statistics, seasonal encodings, and climate-derived indices) for

characterizing the delayed and non-linear associations between weather variables and dengue incidence.

Comparing several machine learning models for dengue prediction within time-aware validation settings to prevent data leakage and provide realistic performance estimation when predicting outbreaks.

To select the best ensemble model for spatio-temporal dengue forecasting and explore its performance at different temporal horizons and administrative levels.

To utilize explainable artificial intelligence (XAI) tools on the best performing ensemble model post selection to interpret prediction behavior and quantify feature importance for alignment with existing epidemiological understanding.

To evaluate whether the new forecasting model would be feasible to use as an operational tool for dengue surveillance and early warning systems in supporting the planning of public health interventions strategies, specifically in Bangladesh.

1.5 Thesis Contribution

There are several major contributions that this dissertation has contributed to the field of data-driven infectious disease forecasting, particularly dengue outbreak prediction in Bangladesh. The main contributions of this work can be summarized as:

- A division-wise spatio-temporal framework for dengue forecasting in Bangladesh: To the best of our knowledge, this study is the first to introduce a complete modeling structure that captures temporal dependency and spatial heterogeneity in all 8 administrative divisions of Bangladesh [1] in one unified model, which has been somewhat neglected by existing studies.

- Building an early warning system using a high-quality integrated epidemiological–meteorological dataset: Through daily linkage of dengue surveillance data and high-resolution meteorological reanalysis (data) [9], the research is providing a rigorously processed spatio-temporal dataset that can be used to develop advanced machine learning–based forecasting, addressing challenges around missing values, temporal misalignment and data inconsistency.

- Use of sophisticated feature engineering in climate–dengue modeling: The dissertation illustrates how advanced autoregressive lags, rolling-window statistics, seasonal encodings and a blend of climate indices can be used to model the delayed and non-linear relationship between climatic features and dengue prevalence [16].

- Identification and rigorous evaluation of an ensemble machine learning model: In this work we systematically assess the performance of 18 diverse ensemble-based models [Appendix A], including lightGBM [22], and analyse how (average) predictive performance change according to geography and over time for spatio-temporal dengue forecasting in Bangladesh.

- Incorporation of explainable artificial intelligence (XAI) in dengue prediction: Utilizing SHAP-based explanation techniques for the most accurate ensemble model, this study improves transparency and interpretability, thereby allowing understanding of the relative importance of climatological, temporal and epidemiological drivers in dengue outbreaks.

- Actionable insights for public health decision-making: Aside from the predictive performance, the new approach presents interpretable and district-specific insights which can guide dengue surveillance, early warning systems, as well as evidence-based intervention planning by public health authorities in Bangladesh.

Taken together, these contributions raise the state-of-the-art in spatio-temporal dengue prediction through the integration of ensemble machine learning, robust evaluation and explainability in a context-sensitive framework for Bangladesh.

1.6 Thesis Outline

The rest of this thesis is organized as follows:

- Chapter 2 presents a comprehensive literature review of related work.
- Chapter 3 describes the methodology and research design.
- Chapter 4 presents the experimental results and analysis.
- Chapter 5 Interprets model predictions using explainable AI techniques.
- Chapter 6 concludes the thesis and discusses future work directions.

Chapter 2

LITERATURE REVIEW

2.1 Traditional Epidemiological Models for Dengue

Existing studies in dengue outbreak prediction have so far mainly focused on classical epidemiological and statistical modeling methods [18]. Such models attempt to capture the transmission of diseases with prespecified mathematical forms and assumptions, typically representing processes at a population scale among human hosts, mosquito vectors and environmental factors [13].

The compartmental modeling framework, including the Susceptible–Infected–Recovered (SIR) and Susceptible–Exposed–Infected-Recovered (SEIR) models are some of the most commonly used methods in dengue epidemiology [18]. These models describe the progression of disease through a number of compartments that are represented by differential equations and have previously been used in the context of dengue for transmission dynamics, reproduction numbers, and intervention scenarios. Extensions of these model account for vector population, seasonality forcing and climate dependent parameters to more accurately capture mosquitoes mediated transmission dynamics [16]. Although helpful in theory, actually forecasting with such models is hindered due to strong homogeneous-mixing assumptions and fixed parameterization [17].

Time-series models are also commonly used for dengue predictions. Methods such as autoregressive integrated moving average (ARIMA) model and its seasonal extended form are employed to capture temporal trends and seasonality in dengue incidence data. Now some work have further developed these models, adding exogenous climatic variables giving ARIMAX kind of expressions which include temperature and rainfall impact. While such models may be able to capture reasonably well predictive trends at short leads, the assumption of linearity in these models does not allow for a good representation of the complex, non-linear dengue–climate interaction and lagged effects [13].

In Bangladesh and the South Asian subcontinent, a number of studies have used regression-based, compartmental models as well to explore patterns in dengue transmission and

determinants at seasonal scale. These works have provided valuable computer-based regional perspectives, especially as related to monsoon rainfalls and temperature variability [13]. But these work make focus on some cities or any periods, and do not reveal the spatial heterogeneity among administrative divisions and temporal dependency for long time yet.

An inherent limitation of conventional analytical and statistical modelling applied in an epidemiological framework is a lack of flexibility when these types of data (e.g. high dimensional and non-linear interactions) are considered. Since the transmission of dengue is driven by multiple interacting factors that operate across different temporal scales, inflexible model structures may suffer from a lack of fit and low generalization performance, particularly when data are abundant [17]. In addition, the inference of parameters in compartmental models usually relies on strong assumptions or cross-calibration with external data that may not be easily applied to other regions with different socio-environmental conditions [18].

Yet, despite such limitations, classical models are an essential benchmark for interpreting dengue dynamics as to date and also the more advanced data-driven approaches [13]. Their limitations, however, have sparked the increasing popularity in machine learning (ML)–based forecasting approaches that are more effective for detecting non-linear spatio-temporal structures in large-scale epidemiological datasets. This shift to data-driven modeling sets the stage for the rest of this review.

2.2 Machine Learning in Infectious Disease Forecasting

Recently, machine learning (ML) models have received widespread attention in infectious disease prediction [15] for their capacity to learn intricate non-linear relationships from data. In contrast to classical epidemiological and statistical models, most machine learning approaches do not need to rely on strong prior assumptions about the disease transmission mechanism and are thus particularly appropriate in a data-rich system considering several interacting factors impacting disease dynamics.

When machine learning was first introduced to infectious disease forecasting, regression based and shallow learning methods (e.g., trees, support vector machines, artificial neural networks) were predominantly used for the purpose of predicting case incidence with historical case reporting and weather data [37]. These works showed the potential of ML

models to do better than classical linear models by modelling non-linear dependencies and interaction effects, especially when climatic drivers are determinants.

As large-scale epidemiological and environmental data continue to be more available, statistical based machine learning methods such as ensemble-based techniques have become good tools for disease prediction [33]. Methods such as Random Forests, Gradient Boosting Machines, and Extreme Gradient Boosting (XGBoost) [21] are commonly used in infectious disease prediction problems including dengue, influenza [22],[23] and malaria prediction. Ensemble models group a number of weak learners in order to improve the generalization capabilities and reduce the variance, as well as robustness to noisy or heterogeneous data.

The effectiveness of machine learning models for forecasting dengue at various spatial and temporal scales has been demonstrated in several studies. These studies often include lagged disease incidence, temperature, precipitation and humidity as predictors, demonstrating the temporal dependency and climate influence on dengue transmission. Many of such studies, however, have a narrow geographic focus or short-term horizon and use random data splitting which ignores causality to mechanisms causing the outcome, thus leading to over-optimistic performance measures [30].

Within the Bangladeshi and South Asian context, machine learning (ML) dengue forecasting projects are few in number compared to worldwide disease model studies. Another drawbacks of the existing works are that they mainly focus on city detection,or seasonal trend analysis and lack spatio-temporal evaluation across administrative divisions at once [24]. Furthermore, only a fraction of these studies make use of strict time-aware validation procedures that are necessary for the realistic outbreak prediction and operational deployment.

Despite the ability to make good predictions, machine learning models are often accused of being 'black boxes', this aspect is particularly concerning when it comes to applications in high-risk settings such as public health. Sophisticated models, in particular, can act as black boxes: Great predictions without clear explanations for why the predictions were made. This constraint has driven a recent surge of interest in interpretable machine learning methods [32] which seek to trade predictive performance for transparency and trust.

In general, the available research shows that machine learning models can increase the accuracy of infectious disease forecasts. It also simultaneously exposes distinct limitations concerning spatio-temporal generalization, validation stiffness, regional specificity and

interpretability [14],[35]. Formally addressing these gaps and integrating ensemble learning, time-aware evaluations and explainable modeling into a unified forecasting framework, is challenging work that naturally leads to the approach we followed in this dissertation.

2.3 The Role of Meteorological Factors in Dengue Transmission

Weather conditions such as the temperature, rainfall and humidity have significant impact on dengue fever transmission. The Aedes mosquitoes transmitting the virus are affected by these two factors in Bangladesh [17].

The temperature has an impact on the mosquito life-cycle and dengue virus replication. Active development of mosquitoes is favoured by temperature, and mosquito biting rate increases; consequently there is a higher risk of transmission. Outbreaks in Bangladesh are known to be associated with higher temperature and especially during summer season [16].

Creating these habitats is important to support the breeding of Aedes mosquito where stagnant water is required for their larvae to develop. Being a rainy country, the monsoon in Bangladesh usually results in heavy rainfalls which is ideal for mosquito breeding and hence there have been reported increased of dengue cases during or after the rainy season [23].

Humidity is another critical piece of the puzzle: high humidity favors the survival of mosquitoes and the transmission of virus. Mosquitoes are better adapted to dry climates since they lay eggs in small containers and could store water up to 2 weeks for their larvae [17] but humid weather is also ideal for them to prevent dehydration and stay active, hence dengue incidence might further increases [17],[6].

These synergising climate factors contribute to seasonality in dengue transmission, with peak incidence generally reported during monsoon and post-monsoon seasons. It is crucial to understand the role of these environmental factors in order to predict epidemics and undertake timely public health measures.

2.4 Explainable AI (XAI) in Healthcare and Epidemiology

There is growing use of complex machine learning models in health and epidemiology, giving rise to key issues around transparency, trust and accountability. Although state of the art methods like ensemble gradient boosting machines and deep learning models can

outperform traditional logistic regression based methods, they are 'black-box' predictors that lack transparency – a key requirement in high-stake domains including public health decision-making [30]. Accordingly, explainable artificial intelligence (XAI) has become a critical requirement of contemporary data-driven healthcare applications.

Explanation models can be defined as those that grant human interpretable explanations for a model's predictions by quantifying the input feature contributions to model outputs[20]. In epidemiological modelling, interpretability is an obvious concern given that modelled outputs should accord with biological plausibility and well-documented means of disease transmission [3]. In the absence of interpretability, public health organizations will hesitate to act on model outputs for intervention planning even in the face of high predictive accuracy.

Multiple explainability techniques have been described in the literature, including model-specific methods such as feature importance of tree-based models and model-agnostic ones used as partial dependence plots (PDPs), local interpretable model-agnostic explanations (LIME) or Shapley Additive explanations (SHAP) etc. Of them, SHAP has been widely used because it is strongly theoretically-based on cooperative game theory, and can offer both global and local interpretation over various kinds of model classes [20].

Within the field of infectious disease forecasting, XAI methods have recently been applied to interpret machine learning models that forecast the spread of diseases like dengue, influenza and COVID-1927. These analyses show that explainability tools can be used to understand the crucial climatic, temporal and demographic drivers of disease transmission, examine how model results compare with available epidemiological knowledge and potential sources of biases or artifacts in the data. These can be crucial in order to prevent model predictions from being not only accurate, but scientifically realistic and actionable [30].

In dengue forecasting work explainable models have been applied to find the importance of lagged incidence, temperature, rainfall and seasonal variables. XAI studies have demonstrated that the feature contributions are frequently different from one region to another, and dynamics in time, thus highlighting the need for spatio-temporal modeling and region-specific interpretation. But most of previous work only plays with the explanation and not really up to solving, or consider small-area scope, much can be done on big-picture interpretability.

In spite of the increasing popularity of XAI, there is a limited systematic application to time-varying dengue forecasting contexts that include spatial information—especially in the context of South Asia. There is a gap in the literature for methods, which leverage high performing ensemble models with robust explainability analysis that can be used to interpret

results transparently and achieve credible, policy-relevant disease forecasts [30]. Satisfying this need is of great value in countries like Bangladesh where public health decisions have to rely on understandable evidence that are based on epidemiology and climate conditions at a local and national scale.

In conclusion, XAI is of utmost importance to narrow the chasm between predictive accuracy and operational applicability among machine learning models for epidemiology. By allowing for the interpretability of ensemble model predictions, XAI instils trust, aids in scientific validation and assists with informed public health decision-making [12]. These reasons literally justify the use of SHAP-based explainability methods in this thesis to interpret spatio-temporal dengue outbreak predictions.

2.5 Research Gaps in the Context of Bangladesh

Despite significant strides in dengue epidemiology and forecasting research at the international level, there are still a number of knowledge gaps when focusing on these methods as it relates to the Bangladesh context. While the existing studies (both traditional and machine learning based) have provided useful insights on dengue transmission dynamics, a majority of these studies lack considerations to address the specific spatio-temporal, climatic and operational constraints encountered by dengue surveillance systems in Bangladesh.

First, there is a significant spatial discrepancy in terms of the covered history. The majority of dengue prediction studies carried out in Bangladesh, as well as South Asia regionally, have concentrated on single cities or smaller geographic areas; usually a city centre. Such focused analyses completely miss inter-division variation in the dynamics of outbreak due to differences in climate, population density, urbanization patterns and healthcare infrastructure. Therefore most of the models are not generalizable at national level and inapplicable for division wise early warning setup [38].

Second, most previous work has shallower time depth and less validation strength. Short training periods and random train–test splits are still widely reported in the literature, yielding performance figures reflecting situations not likely to occur when forecasting. For dengue, the prediction of outbreaks require prospective forecasting that is time dependent -- a fundamental limitation in the absence of time aware evaluation methodologies [37]. This discrepancy challenges the feasibility of practical reported forecasting models.

Third, although machine learning and ensemble models are considered promising in terms of predictive performance, their systematic use for spatio-temporal dengue forecasting in Bangladesh is still limited. Advanced feature engineering, Long-term division-wise data and ensemble learning are few of the studies come together under a single platform [34]. Additionally, the actual advantages of ensemble boosting models over them can be overwhelming indicated in realistic validation scenarios.

Fourth, the lack of model interpretability is insufficiently addressed in regional dengue forecasting literature. While some studies present feature importance scores, an in-depth explainability analysis that gives Global and Local explanation and an understanding of model behaviour is uncommon [32]. This restriction significantly undermines the practical usability of such forecasting models for public health decision-making, as health authorities need clear and interpretable explanations in order to evaluate drivers behind an outbreak and interventions to justify.

Finally, there is a lack of comprehensive models with consideration of spatial heterogeneity, temporal dependency, rigorous evaluation, and explainability based on high-resolution epidemiological and meteorological data adapted specifically to Bangladesh. Existing works often treated these aspects separately without incorporating them in an end-to-end modeling pipeline.

The literature review we conducted clearly shows that there are research gaps, in the context of Bangladesh, in terms of spatio-temporal coverage over nationwide scope, time-aware validation and ensemble-based modeling as well as explainability. Filling these missing links would be the development of a forecasting model that exploits division-wise data, sophisticated feature extraction techniques, ensemble learning and explainable artificial intelligence. The abovementioned gaps specifically inform the methodology and empirical orientation of the current thesis.

Chapter 3

METHODOLOGY

3.1 Introduction

This chapter presents the methodological framework of dengue outbreak prediction in Bangladesh from surveillance data specially focusing on the process of data collection, integration and preparation. Accurate forecasting of dengue incidence depends on the availability of high quality epidemiological data integrated with environmental covariates whose variation affects the dynamics of disease transmission. To bridge this gap, we match reported dengue surveillance data with high quality reanalyzed meteorological fields in time and space to define the spread of dengue. The emphasis is on the reliability, temporal consistency and spatial coherence of data across sensors. In addition this chapter introduces the procedures of exploratory data analysis applied to study statistical properties, seasonality and regional variation in dengue counts. Data sparsity, zero-inflation, missing values and reporting inconsistencies are some of the major challenges that are systematically dealt with using pertinent preprocessing methodologies [17]. The method steps in this chapter have created a strong and analysis-performing dataset, which can be used for more advanced feature engineering and machine learning modeling that follows.

In this study a structured and time-sensitive machine learning framework is used for dengue incidence prediction. The proposed method is aimed at robustness, temporal consistency as well as reproducibility across the modeling steps [33]. The first step consists in the gathering of dengue surveillance and meteorological datasets from trusted sources followed by a consistent data cleaning process that includes filling missing values, inconsistencies and temporal misalignment.

This is then followed by more sophisticated feature generation methods, such as the creation of lagged variables to model the temporal correlation known in dengue transmission dynamics [13],[16]. To avoid data leakage and maintain the causality of the time-series data, only historic information is used to compute features. The dataset is then split using a strict-time series form of chronologically based hold-out (i.e., so that model training and testing reflect real world forecasting).

Several ML approaches are used to develop and compare the baseline performance of models, through analyzing parameters that leads to their better predictive performance by randomized search and cross-validation [37],[22]. The best model is retrained on the entire training period and tested for novelty on a future unseen test set. An overview of the entire methodological process implemented in this study is depicted in **Figure 3.1**, describing the pivotal phases ranging from data preprocessing to final performance assessment.

Several machine learning models are utilised to achieve baseline performance, and then hyperparameter tuning via the random search cross-validation method for better prediction accuracy [21],[22]. The best model is re-trained with the complete train period and evaluated on a unseen future test set. The eclectic approach employed in this study is depicted in **Figure 3.1**, where every major phase from data preprocessing to overall validation effectiveness is briefly outlined.

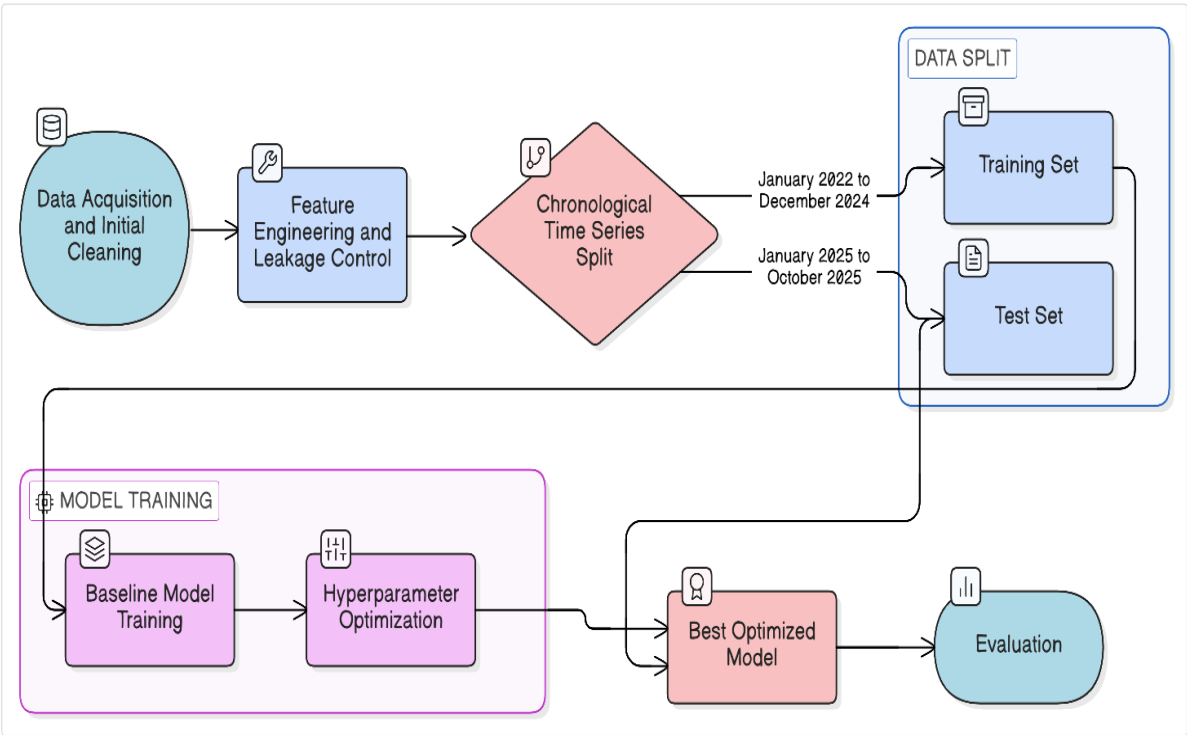

Figure 3.1: Overall Methodology of Our Work

Figure 3.1 Logic overview of the proposed method in dengue prediction. The figure provides an overview of the end-to-end pipeline, including data retrieval and processing, feature generation with leakage control, chronological train–test partitioning, baseline

model training and hyperparameter tuning (opt) along with final evaluation by a comprehensive list of evaluation scores.

3.2 Data Sources

In this paper, the epidemiological surveillance data is complemented with high resolution meteorological data to investigate dengue transmission dynamics in Bangladesh [5]. All data were collected and analyzed with reproducible, programmatically generated scripts [8] to ensure transparency, consistency and reproducibility throughout the pipeline.

3.2.1 DGHS Daily Dengue Patient Records

Epidemiological data The main epidemiological information included the counts of dengue cases reported in a daily basis filed by Directorate General of Health Services (DGHS) from Bangladesh [5]. They are the official division-based dengue case statistics and represent the target variable for model building and evaluation [23]. The data set includes all eight administrative regions for national spatio-temporal analysis. The DGHS records on dengue are a frequently applied data source for dengue research in Bangladesh considering its authoritative status and standardized reporting form [40].

Like most surveillance real-world dataset, the DGHS records are plagued with missing information, reporting irregularities and long spates of zero incidence over different divisions. These traits call for cautious exploration and preprocessing reflects before any predictive modelling is performed [1],[2].

3.2.2 Meteorological Data from Script-Based Retrieval (Open-Meteo and ERA5)

Epidemiological data The main epidemiological information included the counts of dengue cases reported in a daily basis filed by Directorate General of Health Services (DGHS) from Bangladesh [5]. They are the official division-based dengue case statistics and represent the target variable for model building and evaluation [23]. The data set includes all eight administrative regions for national spatio-temporal analysis. The DGHS records on dengue

are a frequently applied data source for dengue research in Bangladesh considering its authoritative status and standardized reporting form [40].

Like most surveillance real-world dataset, the DGHS records are plagued with missing information, reporting irregularities and long spates of zero incidence over different divisions. These traits call for cautious exploration and preprocessing reflects before any predictive modelling is performed[1].

Table 3.1: Summary of Data Sources Used in the Study

Data Type	Source	Temporal Resolution	Spatial Resolution	Description
Dengue Incidence	DGHS, Bangladesh	Daily	Administrative division	Official reported dengue cases
Meteorological variables	Open-Meteo / ERA5	Daily	Gridded (aggregated)	Temperature,rainfall, humidity, climate indices

3.3 Dataset Characteristics and Exploratory Data Analysis

The statistical properties of the integrated dengue–meteorological dataset and exploratory data analysis attempts to evaluate its temporal dynamics, spatial variability, and distributional nature is described in this section. Exploratory data analysis (EDA) yields invaluable information that contribute to the feature engineering, and preprocessing processes in subsequent phases [13].

3.3.1 Dataset Overview

Upon integration and alignment, this results in 11,088 daily observations between 15 January 2022 to 31 October 2025 for each administrative division. The data set defines a multivariate spatio-temporal time series that includes epidemiological, climatic, seasonal and engineered temporal features [34].

Table 3.2: Overview of the Integrated Dengue–Meteorological Dataset

Attribute	Description
Spatial units	8 administrative divisions of Bangladesh
Temporal resolution	Daily
Target variable	Dengue case count (Patients)
Climatic variables	Temperature, rainfall, humidity, derived indices
Time span	15 January 2022 – 31 October 2025
Total observations	11,088
Data structure	Multivariate spatio-temporal time series

3.3.2 Temporal Distribution and Zero Inflation

Exploration of the data shows marked temporal seasonality and inter-annual fluctuations in dengue cases, showing distinct peak outbreaks during monsoon and post-monsoon periods. In off-peak periods one observes long times of low/zero incidence in various compartments [13].

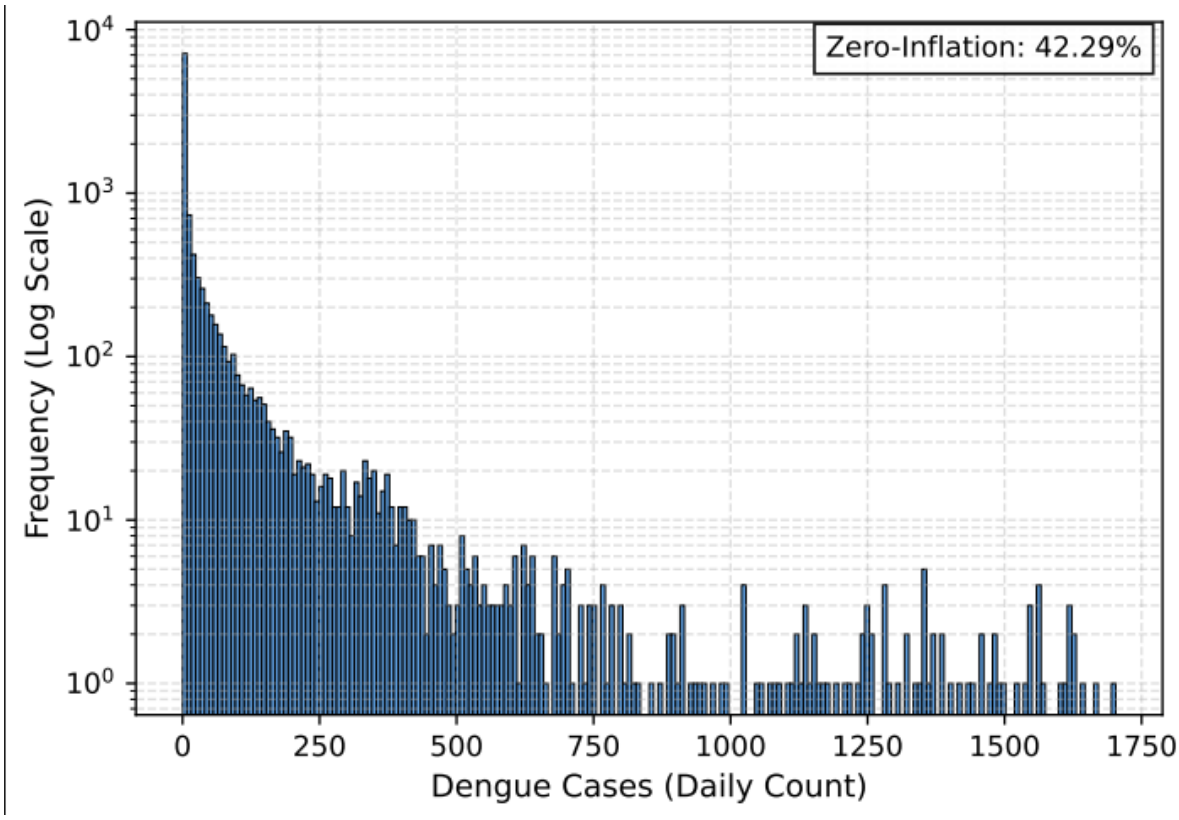

Figure 3.2 Distribution of daily dengue cases showing raw and log-scaled views, highlighting skewness and zero inflation.

The distribution of daily dengue cases throughout the study is shown in **Figure 3.2**, with which we highlight complexity due to the epidemiological data nature. The following important attributes are noted:

- **Zero-Inflation:** A large proportion of the data, **42.29%** precisely, were zero case days. This is because dengue in Bangladesh is circumscribed by seasons, with little transmission occurring during the dry winter months.
- **Very Skewed Distribution:** The data has extreme right-skew (positive skew). Even if most case counts are clustered between 0 and 250, there is a long tail with more than 1,500 cases that corresponds to intense outbreak periods.
- **This logarithmic scale:** is used to maintain visibility of infrequent but high-impact occurrence days and prevent the dominate of low-case days.
- **Modeling Interpretation:** The high proportion of zeros and very large outliers indicate that the data is not normally distributed. This is what justifies the use of non-linear, ensemble machine learning models such as **LightGBM** or **XGBoost** that can handle these distributions well.

The daily dengue case range is strongly right-skewed and zero-biased. Roughly 42.29% of daily reports were zero cases, with epidemic (outbreak) periods yielding outliers greater than 1,500 cases per day. This distributional behaviour suggests the use of variance-stabilising transformations before modelling.

3.3.3 Spatial Variability Across Administrative Divisions

There is considerable spatial heterogeneity across the eight administrative divisions of Bangladesh. More urban divisions (chiefly Dhaka) have a much higher mean incidence and outbreak magnitude [40] with other divisions more sparing in their number of cases.

Table 3.3: Division-wise Summary Statistics of Dengue Incidence (2022–2025)

Division	Mean	Median	Maximum	Standard Deviation
Barishal	48.29	6.0	477	84.81
Chattogram	57.34	12.0	547	96.32
Dhaka	217.07	39.5	1703	358.25
Khulna	37.24	2.0	436	79.12
Mymensingh	10.49	1.0	105	19.21
Rajshahi	21.49	0.0	246	44.66
Rangpur	5.79	0.0	88	14.05
Sylhet	1.53	0.0	30	3.97

These numbers reveal strong spatial inequalities and motivate the Div-Wise STMM.

3.4 Data Cleaning and Pre-processing

This subsection describes the thorough pretreatment methods that was performed on the data before model training. Special focus is given to account for zero-inflation in dengue reports as well as preserving temporal causality and avoiding information leakage. It is essential that the ensemble models accurately represent dengue virus' complicated transmission dynamics

in Bangladesh while avoiding both bias and temporal inconsistency that can be detrimental to predictive accuracy [17],[23].

3.4.1 Handling Missing Values and Data Consistency

The dataset was relatively complete; nevertheless, one gap in the data was discovered between **July 18 to 23 July,2024**. This aberration resulted from a country-wide network and communication blackout in Bangladesh [5] that temporarily interruptedofficial disease surveillance and meteorological broadcasting.

To preserve continuity in the time series, missing values for this period were estimated by division-wise monthly average imputation [23]. This method guarantees the epidemiological idiosyncrasies of the eight administrative regions are maintained, while avoiding "look-ahead" bias or information leakage during training. TT and forward fill was used to account for smaller discrepancies elsewhere in the dataset preserving the sequence of events [17].

3.4.2 Target Variable Transformation

As dengue case counts are highly skewed and zero-inflated, we log transformed the data to stabilize variance and enhance modeling learning [21],[22]. The transformation used is:

$$y' = \log(1 + y)$$

where , is the original daily dengue case. This transformation diminish the effect of outliers, and provides more consistent variance throughout the time series, which is required for models like XGBoost and LightGBM which are not tolerant to such high skewness levels.

In test and result interpretation, predictions went back to original scale using:

$$\hat{y} = \exp\,(y') - 1$$

This is to maintain interpretability of performance measures and predictions in an epidemiological context. By doing so, model predictions are adjusted to the real-world dengue incidence values, thus facilitating public health authorities to appropriately interpret and use the results when making decisions.

3.4.3 Temporal Integrity and Leakage Control

All lagged and rolling features were engineered forward-aggregations into the present day rather than backward shifts through time [18]. This makes sure that future data do not influence the present predictions, therefore preserving the temporal causality characteristic of time-series data. We used date-time specific splitting with years 2022 to 2024 for training and 2025 till October for out-of-sample evaluation [30]. This train-test split adheres to time-series forecasting best practice in that no future data is used to predict past or present events, namely preserving the chronology of dataset [37].

This cautious treatment of time series data makes our model less prone to overfitting and allows us to evaluate its performance in real-world situations, where predictions are made relying on historical data, not what will happen in the future.

3.4.4 Feature Scaling

Continuous input features were standardized to have 0 mean and unit variance based on the statistics of the training data only. The same scaler was applied to the test data to maintain consistency and prevent information leakage. The preprocessed dataset was further prepared for model training and evaluation in the following sections [1],[2].

3.5 Advanced Feature Engineering

This part describes how advanced dynamical patterns are learned to induce these latent dynamics. By mapping raw data into a high-dimensional feature space—accounting for temporal lags, seasonal harmonics, and derived climate indices—we allow ensemble learners to assess fine-grained precursors to dengue epidemics [6]. These high-level features are

essential to model the strong non-linear, spatio-temporal relations that characterize the epidemiology landscape in Bangladesh [16].

3.5.1 Target Variable Transformation (Log-Scaling)

The dengue case count data is highly skewed and zero-inflated [17]. To account for this, we take the logarithm of the response to stabilize the variance and diminish the effect of extreme outliers [21],[22]. The transformation formula is:

$$y_t' = \log(y_t + 1)$$

Where:

- y_t is the original dengue case count at time t,
- y_t' is the transformed target variable

This transformation scales the data more robust for machine learning models.

3.6 Temporal Feature Extraction

The time-dependent patterns and delayed effects in dengue incidence data are extracted by temporal feature extraction [13],[16]. As dengue transmission rates are impacted by historical exposure and environmental effects, lag terms for the case counts for risk z and for weather conditions r are constructed [1],[2]. These lagged features enable the models to account for the delayed influence of climatic variables and recorded dengue cases at previous timepoints on future outbreaks [14].

Moreover, moving averages of weekly (7-day) and biweekly (14-day) coverage are estimated to dampen short-term variability and therefore bring out longer term trends in dengue cases. Time factors such as day of the year are included to account for seasonal patterns often seen in dengue transmission [34]. All temporal features are generated from historical information without considering any current observations, preserving the temporal sequence of observations and avoiding contamination problems that may cause unreliable dengue forecasting.

3.6.1 Autoregressive Lags (Patients_lag)

In order to incorporate temporal information of the recorded data, autoregressive lag features [13] are added. These are the previous day's values of dengue case counts, and they are defined as follows:

$$y_t^{\text{lag}(n)} = y_{t-n}$$

Where:

- $y_t^{\text{lag}(n)}$ is the dengue case count from $n$days before time t.

3.6.2 Rolling Window Statistics

Rolling window statistics are used to capture short-term variations in dengue outbreaks [19]. But the for rolling mean is:

$$\text{mean}_t^w = \frac{1}{w} \sum_{i=t-w+1}^{t} y_i$$

Where:

- mean_t^w is the rolling mean of dengue cases over a window of size w,
- y_i is the dengue case count at time i.

Such rolling statistics will provide information on trend for dengue outbreaks.

3.7 Seasonal Encoding (Fourier Series and Harmonic Regression)

The Fourier series [16] is applied in order to model the seasonal component of dengue outbreaks. The Fourier transform over a period is:

$$S_t = \sum_{k=1}^{K} [\, A_k \cos\left(\frac{2\pi kt}{T}\right) + B_k \sin\left(\frac{2\pi kt}{T}\right)]$$

Where:

- A_k and B_k are Fourier coefficients,
- t is time,
- T is the period of the cycle (e.g., a year for seasonal patterns).

This enables the reproduction of seasonal variations in the force of infection of dengue.

3.8 Climate Indices (Heat Index and Rain–Temp Index)

We will model the influence of temperature and humidity on dengue outbreaks by the means of two climate indices [11],[17], which are:

Heat Index (HI):

$$HI = c_1 + c_2 T + c_3 H + c_4 TH + c_5 T^2 + c_6 H^2 + c_7 T^2 H + c_8 TH^2$$

Where:

- T is temperature in °C,
- H is humidity (%),
- $c_1, c_2, \ldots, c_8$ are constants.
- **Rain-Temperature Index (RTI)**:

$$RTI = \text{Rainfall} \times \text{Temperature}$$

These indicators facilitate capturing the environmental determinants of mosquito breeding and dengue transmission escalation.

Table 3.4: Feature Engineering for Dengue Outbreak Forecasting

Feature Category	Feature Name(s)	Engineering Method	Purpose / Rationale
Target Variable	Patients	Raw daily dengue case count (later log(1+Patients) in modeling)	Captures outbreak intensity; log-transform stabilizes variance and reduces skewness
Core Weather Features	max_temp, min_temp, avg_temp, temp_range, humidity, humidity_norm, rainfall, rain_binary	Direct extraction, normalization, binary encoding	Weather strongly influences mosquito breeding and virus transmission
Lag Features	Patients_lag3, Patients_lag7, Patients_lag14; avg_temp_lag3, avg_temp_lag7, avg_temp_lag14; rain_binary_lag3, rain_binary_lag7, rain_binary_lag14; humidity_norm_lag3, humidity_norm_lag7, humidity_norm_lag14	Time-series lagging (3, 7, 14 days)	Captures delayed effects of climate and past cases on current outbreak risk
Rolling Window Features	rolling_Patients_7d_mean, rolling_Patients_14d_mean; rolling_avg_temp_7d_mean, rolling_avg_temp_14d_mean; rolling_rain_binary_7d_mean, rolling_rain_binary_14d_mean; rolling_humidity_norm_7d_mean, rolling_humidity_norm_14d_mean	Moving averages over 7 and 14 days	Smooths short-term fluctuations; highlights medium-term climate and case trends
Derived Climate Indices	temp_humidity_index, rain_temp_index, rain_humidity	Composite indices combining multiple weather variables	Encodes nonlinear interactions (e.g., high humidity + high temp = favorable mosquito growth)

Seasonal Features (One-Hot)	season_Winter, season_Pre-Monsoon, season_Monsoon, season_Post-Monsoon	One-hot encoding of categorical season labels	Captures abrupt seasonal shifts in dengue dynamics not explained by smooth Fourier terms
Seasonal Features (Fourier)	fourier_sin, fourier_cos (optional: fourier_sin2, fourier_cos2)	Harmonic regression using sine/cosine transformations of day-of-year	Models smooth cyclic seasonality in dengue outbreaks
Spatial Features	division_Barishal, division_Chattogram, division_Dhaka, division_Khulna, division_Mymensingh, division_Rajshahi, division_Rangpur, division_Sylhet	One-hot encoding of geographic divisions	Enables one global model across divisions while preserving regional heterogeneity
Temporal Features	day_of_year	Calendar-based numeric encoding	Provides temporal context for seasonality and Fourier transformations

3.9 Experimental Setup and Time-Series Train-Test Split (2022–2025)

The dataset was partitioned into training and test sets according to the temporal interval.

- **Training data:** January 2022 to December 2024
- **Testing period:** January 2025 until October 2025

The time series train-test split is to simulate real-world prediction settings and appropriate for a predictive modeling study [18].

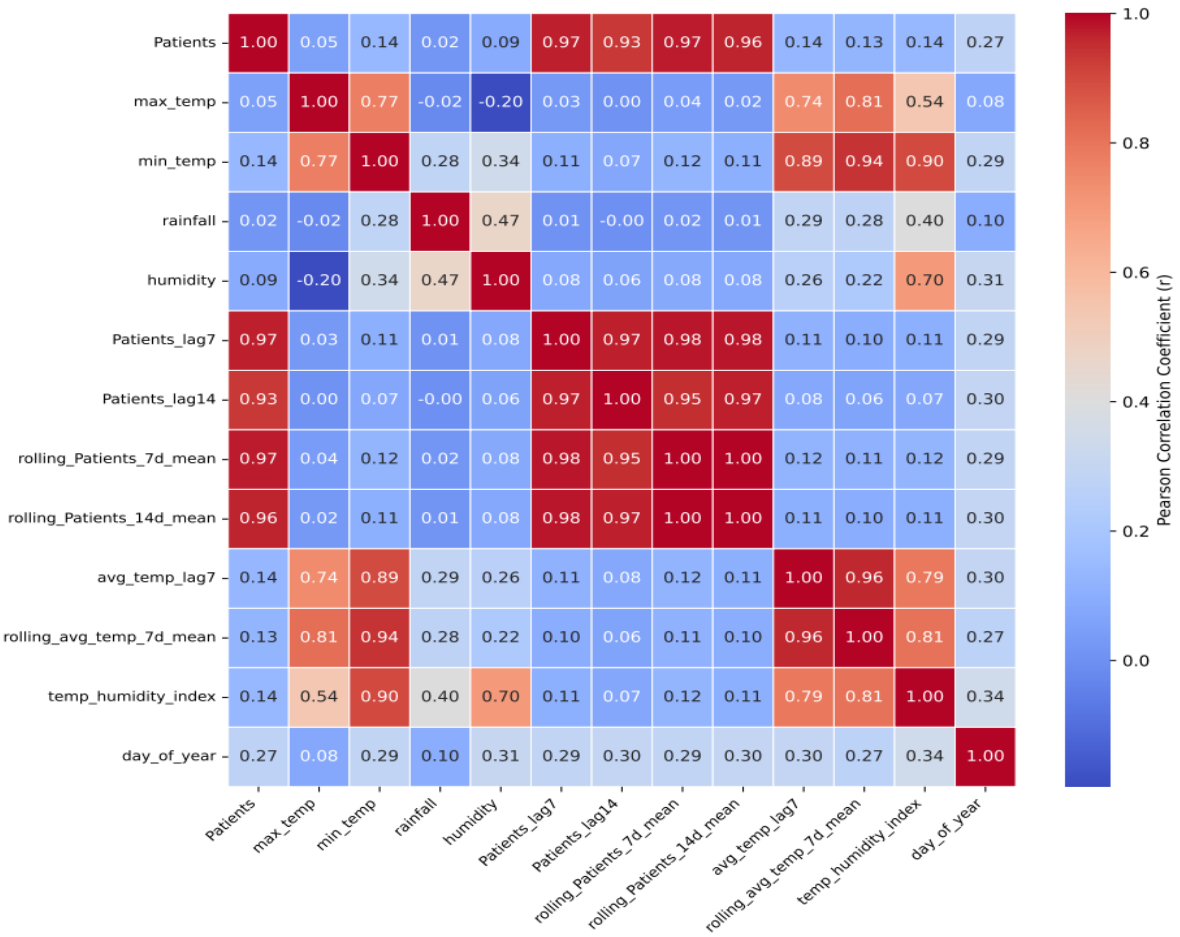

Figure 3.3: Correlation Heatmap

Figure 3.3 illustrates the pearson correlation heatmap among dengue case counts and the chosen time-based and meteorological features is plotted as figure 3.3 Dengue cases show strong positive correlations with their lagged and rolling covariates, which suggest the presence of temporal dependence in dengue incidence. The meteorological factors temperature, humidity and rainfall based on their midpoint values have only moderate correlation with dengue cases indicating their influence in dengue transmission dynamics. For the (b) and (c) plotting features, the heatmap shows interesting relations between variables and can be used in selecting the important characteristic features of the dengue forecasting models.

Correlational analyses between dengue incidence and lagged variables and climatic predictors reveal strong auto-correlations in dengue case counts, and comparatively weak instantaneous linear relations with raw meteorological data. This supports inputting lagged and rolling features through feature engineering.

3.10 Modeling and Optimization

With the dataset pre-processed and spatio-temporal, and feature space constructed, this section narrates the modeling aspect of this study. In order to model the complex, non-linear relationships between climatic predictors and dengue occurrence, we assess a wide range of machine learning methods [6],[7],[15].These all vary between linear benchmarks and complex ensembles that are able to deal with high-dimensional data and regional heterogeneity [33].

Specifically, the aim is to identify a model that is both accurate in prediction and robust enough for a nationwide early warning system. Every candidate model is fine-tuned through an intensive optimization to fulfil high performance over the eight administrative division of Bangladesh [1],[2].

3.10.1 Selection of Machine Learning Models

We have selected the following six machine learning models for this study [21],[22]:

- **Linear Baseline:** Ridge Regression (Sec 3.10.1)
- **Boosting Ensemble Models:** XGBoost, CatBoost and LightGBM (3.10.2)
- **Bagging and Kernel Models:** Random Forest and SVR(Section 3.10.3)

These models were chosen to evaluate various approaches, spanning from linear to more complex ensemble-based models.

3.10.2 Linear Baseline: Ridge Regression

The ridge regression is a linear model applied as the baseline. It imposes L2 regularization on the linear regression equation, a technique that can prevent overfitting when large coefficient values are penalized [18]. This is especially helpful when there may be multicollinearity between features in the data set.

The model is built on the log-transformed dengue counts and compared with several more complex models in terms of how much variance they explain [19].

3.10.3 Ensemble Boosting: LightGBM, CatBoost, and XGBoost

Ensemble models such as LightGBM, CatBoost and XGBoost are selected because of their capability of dealing with large scale datasets, learning complex non-linear features and achieving better prediction [21],[22].

- **LightGBM:** A gradient boosting framework that uses a leaf-wise growth algorithm, has faster training speed and better performance in large data [21].
- **CatBoost:** A gradient boosting model that is designed to work with categorical data. It works well with categorical features and is accompanied by regularization tools to avoid overfitting [22].
- **XGBoost:** One of the most effective and widely used gradient boosting frameworks with outstanding performance on structured data problems due to its scalability [22].

3.10.4 Bagging and Kernel Models: Random Forest and SVR

- **Random Forest:** A bagging model combining multiple decision trees result in the aggregate. The practical benefits: This tends to increase prediction accuracy through reduced variance over single DTs [33].
- **Support Vector Regression (SVR):** A non-linear modelling technique that is based on a kernel and captures complex relationships between X and Y. For complex patterns in the data, the Radial Basis Function (RBF) kernel is employed [37].

3.11 Hyperparameter Optimization (HPO) Strategy

To optimize the predictive performance and generalize the models, a structured Hyperparameter Optimization (HPO) system was employed [30]. The approach involved a RandomizedSearchCV with 5-fold cross-validation, which can afford efficient yet fairly extensive search in the high-dimensional parameter space across 100 iterations each model [39].

3.11.1 Log-Domain Optimization for RMSLE

The optimization approach was meticulously designed to tackle the exponential growth patterns and zero-inflated characteristics of the target dataset [17].

- **Target Transformation:** A log transformed target variable was constructed for all models by taking, computed as: log(y + 1) [21].
- **Metric Alignment:** By minimizing the square error in logarithmic domain, HPO method directly optimizes for RMSLE. This guarantees that the search recovers sets of parameters in a relatively balanced sense for both low coverage periods and high growth spike [22].

3.11.2 Model-Specific Search Space Design

The HPO procedure relied on tailored probability distributions to draw hyperparameters according to the mathematical needs associated with individual algorithms:

- **Gradient Boosting Models (LightGBM, XGBoost, CatBoost):** The search was on the compromise between learning rates [0.005–0.15], tree depth and leaf-wise constraints to optimize training speed and performance [21],[22].

- **Ensemble Tree Models (Random Forest):** Hyperparameter, H,u, gave the restriction on feature sampling ratios, e.g.sqrt vs 0.7,min leaf size to minimize correlation between individual trees and control final model variance [33].

- **Kernel-Based and Linear Models (SVR, Ridge):** Regularization strengths (C and Alpha) were tested over orders of magnitude using a log-uniform distribution to sample from the search space (0.1 to 1000). This allows for a tunable search in the hyperplane flexibility and and coefficient shrinkage range[18].

3.11.3 Integrated Constraints and Early Stopping

A key part of HPO for boosting methods, the inclusion of Early Stopping to control overfitting [22].

- **Validation Monitoring:** A separate 10% monitor split was held out from the training data. This split was set aside purely for monitoring the validation error in real-time of each HPO iteration [30].
- **Patience Logic:** A of patience 50 to 100 rounds was enforced. If the validation performance did not improve after setting these rounds, then the current training iteration would end. This dynamic restrict is to make sure that the decision making complex of model may be truly optimum when real convergency instead of a set number of estimators [39].

3.11.4 Final Refitting and Efficiency Analysis

After the best hyperparameter sets were found, a special two-step process was performed to close out the models and evaluate its performance:

1. Comprehensive Dataset Refitting: The best-performing parameters were used to re-instantiate each model and refit the models on the entire training data (2022–2024) [1]. This is to guarantee that our final model include the full variance of available data.

2. Isolated Efficiency Measurement: To be able to fairly compare the computational costs, the time needed for a single fit on the full data set was measured in isolation (recorded after and including four decimal places). This lets the study to have a distinction between Total Search Time (total of. optimization), and Final Training Time (cost for training 339.. the finalized model) [37].

3.12 Model Evaluation Metrics

Four standard statistical metrics are used to objectively evaluate the prediction performance of the models. These include measures of the residuals, that is difference between predicted and observed values from different angles sensitivity to effect from outliers and scale-invariance.

1. Mean Absolute Error (MAE): It measures the average of the absolute errors between predicted and actual values. It returns a linear score in which all individual differences are equally weighted and is therefore a strong indicator of bias on the average model [19].

Equation:

$$MAE = \frac{1}{n}\sum_{i=1}^{n} | y_i - \hat{y}_i |$$

Significance: In dengue forecasting, MAE is important because it measures the average difference between predicted and observed values of dengue cases in real and interpretable units.

The lower the MAE means better accuracy for the model in predicting number of dengue cases needed to plan public health and resources.

Because MAE weighs every prediction error equally, it averages out the forecasting performance comprehensively and is not dominated by large days of disease outbreak.

2. Root Mean Squared Error (RMSE): Root Mean Square Error is the square root of average of squared deviations. Since the errors are squared before they are averaged, RMSE gives a relatively high weight to large errors.

Equation:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

Significance: In predicting dengue, RMSE is essential as it penalizes larger errors in predictions more than the smaller ones. This is highly valuable in detecting ones that does not describe ultrashort surges or outbreaks of the dengue cases. Lower RMSE suggests a

better ability of the model to capture normal shadow and peak transmission, providing valuable information for outbreak preparedness [24].

3. Root Mean Squared Logarithmic Error (RMSLE): For exponential growth or zero-inflation targets, RMSLE is very effective. It focusses on relative error by taking a ratio of the logs of the predicted and actual values.

Equation:

$$RMSLE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\log(\hat{y}_i + 1) - \log(y_i + 1))^2}$$

Significance: RMSLE is a significant improvement over R2 in predicting dengue cases as it considers ratio of the errors instead OF absolute deviation; which suits when number of dengue case counts varies over time * [Explanation by Pina98] It mitigates the effect of large absolute numbers and highlights proportional accuracy, which is relevant to compare predictions during low versus high incidence periods. The lower the RMSLE value is, the better the model works in predicting growth tendency of dengue cases [21],[22].

4. Coefficient of Determination (R^2): The value shows the amount of variance for performance measure variation (dengue cases) that is accounted by the independent variables in model.

Equation:

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Significance: In dengue prediction, R^2 is important because it tells us how well the model being used explains variability in dengue case counts. Higher R^2 means the model can capture the underlying patterns and trends driven by climate or seasonality. This aids in understanding

of the general goodness of fit for the model, but it must be considered along with error-based statistics like MAE and RMSE in order to ensure forecasting performance [33].

Table 3.5: Significance of Evaluation Metrics Used in Dengue Forecasting

Metric	Full Name	What It Measures	Significance in Dengue Forecasting
MAE	Mean Absolute Error	Average absolute difference between predicted and actual dengue cases	Indicates the average forecasting error in real case counts, making it easy to interpret and useful for assessing overall prediction accuracy for public health planning.
RMSE	Root Mean Squared Error	Square root of the mean of squared prediction errors	Penalizes large errors more heavily, making it effective for evaluating a model's ability to capture sudden dengue outbreaks and peak transmission periods.
RMSLE	Root Mean Squared Logarithmic Error	Log-scaled squared differences between predicted and actual values	Focuses on relative prediction accuracy, making it suitable for dengue data with large variations between low and high incidence periods.
R^2	Coefficient of Determination	Proportion of variance in dengue cases explained by the model	Measures how well the model explains overall variability and seasonal patterns in dengue incidence, reflecting goodness of fit.

The table presents the evaluation metrics used to assess the performance of dengue forecasting models. These metrics provide complementary perspectives on prediction accuracy, error magnitude, and overall model fit. Error-based measures such as MAE, RMSE, and RMSLE quantify the deviation between predicted and actual dengue case counts, while R^2 indicates how well the model explains the variability in dengue incidence. Together, these metrics ensure a comprehensive and reliable evaluation of forecasting performance.

Chapter 4

RESULTS AND PERFORMANCE ANALYSIS

4.1 Comparison of Baseline Model Performance

Performance comparison of baseline methods In this section, we compare the performance of other existing baseline models in terms of predicting such dengue outbreaks. The models being evaluated are Ridge Regression, Random Forest (RF), Support Vector Regression (SVR) and all the machine learning ensemble models lightGBM, CatBoost and XGBoost [22]. Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2 (R- squared) and Root Mean Squared Logarithmic Error (RMSLE) [15] are the performance measure utilised for comparison.

The linear Ridge Regression model was limited in its ability to capture the complex, non-linear nature of the relationships present within the dengue data [18]. It served as a comparison baseline, but both accuracy and error rates were comparably worse [19]. Temperature and Cyan Wall Growth Random Forest and SVR fared better when compared to Ridge Regression, [2] however were still no match for more sophisticated models such as LightGBM, CatBoost, and XGBoost that thrived in accommodating the complicated nature of the data due its high-dimensionality [21].

Table 4.1: Baseline Model Performance Comparison

Model	R^2	MAE	RMSE	RMSLE	Training Time
LightGBM	**0.899**	**9.148**	**20.236**	**0.773**	**0.200**
Catboost	0.872	9.852	22.739	0.777	9.222
XGboost	0.849	10.759	24.708	0.812	2.242
RandomForest	0.896	9.173	20.476	0.778	3.458
Ridge	0.640	17.206	38.203	1.097	0.040
SVR	0.689	14.389	35.478	0.957	4.017

As seen in the table, LightGBM and Random Forest are consistent among ensemble models including robust predictions better than traditional ones. One important remark is that in the top left part of this figure (high R^2, low MAE) we can find LightGBM with its low MAE and high R^2, which confirms this algorithm adequacy for dengue prediction.

4.2 Impact of Hyperparameter Tuning on Model Performance

The optimization of hyperparameters was a key point to enhance the performance of the proposed machine learning models [30]. The performance of the models improved significantly based on wide range of seven evaluation metrics by tuning important parameters through RandomizedSearchCV with early stopping [39]. The models especially LightGBM and CatBoost gained the most from hyperparameter tuning with significant decrease in MAE, RMSE and RMSLE [22],[37].

Table 4.2: Tuned Models Performance Comparison

Model	R^2	MAE	RMSE	RMSLE	Training Time
LightGBM	**0.900**	**8.927**	**20.094**	**0.765**	**3.591**
Catboost	0.898	9.026	20.310	0.768	5.797
XGboost	0.891	9.366	20.998	0.772	3.612
RandomForest	0.888	8.984	21.234	0.769	9.684
Ridge	0.637	17.275	38.322	1.097	0.032
SVR	0.383	18.702	49.997	0.910	48.524

Following hyperparameter search, LightGBM and CatBoost performed best with highest prediction accuracy at reasonable training times [33]. We also observed that XGBoost achieved significant improvement in terms of lowering error and better generalization than its performance on the baseline [21].These findings demonstrate that tuning hyperparameters of ensemble models is crucial to reinforce robustness and forecasting performance of ensemble for challenging epidemiological tasks [6],[7].

The improvements in performance after hyperparameter tuning are clearly visible in **Figure 4.1**, which illustrates the percentage changes in MAE, RMSE, R^2, and RMSLE across all evaluated models.

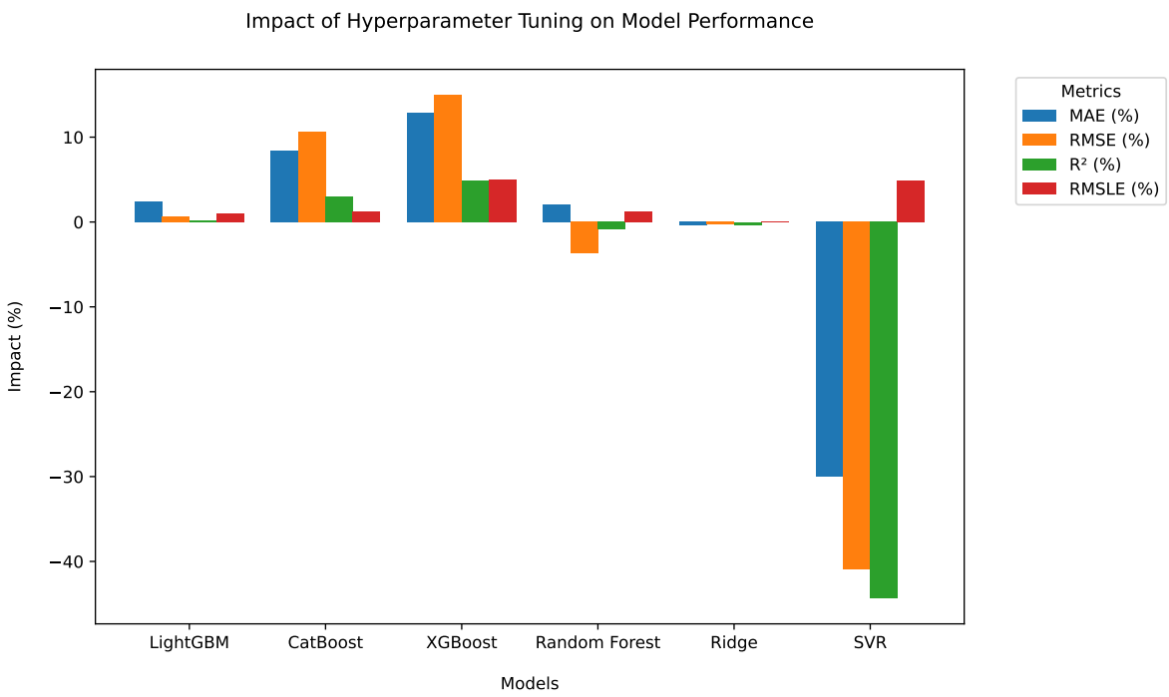

Figure 4.1: Impact on performance after tuning

Ensemble approaches as in with LightGBM, CatBoost and XGBoost show the largest improvements overall with LightGBM well ahead of the others. On the contrary, the Ridge Regression and SVR posed very little change or even negative changes, reflecting their poor applicability in this forecasting problem. These findings show that the accuracy and good general capability of advanced ensemble models can be significantly improved through hyperparameter tuning.

4.3 Analysis of "Accuracy Champion" (LightGBM)

After hyperparameter optimization, LightGBM was determined to be the best-performing model. It obtained a MAE of 8.927, RMSE of 20.094 and an R^2 equal to 0.900 ensuring that it was the best candidate model for forecasting dengue epidemic days outbreak,. LightGBM proved a good ability to handle big and high-dimension data with maintaining precision.

An R^2 value as high as this model's is indicative of a good fit, giving us more confidence in the predictive ability from an epidemiological perspective. Furthermore, its small MAE

and RMSLE also demonstrated the strength of its ability to model complex nonlinear dynamics of dengue transmission within Bangladesh.

Given the better performance measured by a number of evaluation metrics and its efficiency in training, LightGBM is considered the model of choice for operational dengue outbreak forecast deployment.

4.4 Spatio-Temporal Variability in Model Performance

The performance of LightGBM was obtained at various regions such as the eight administrative divisions and five major seasons of Bangladesh: Winter, Spring, Pre Monsoon, Monsoon and Post Monsoon.

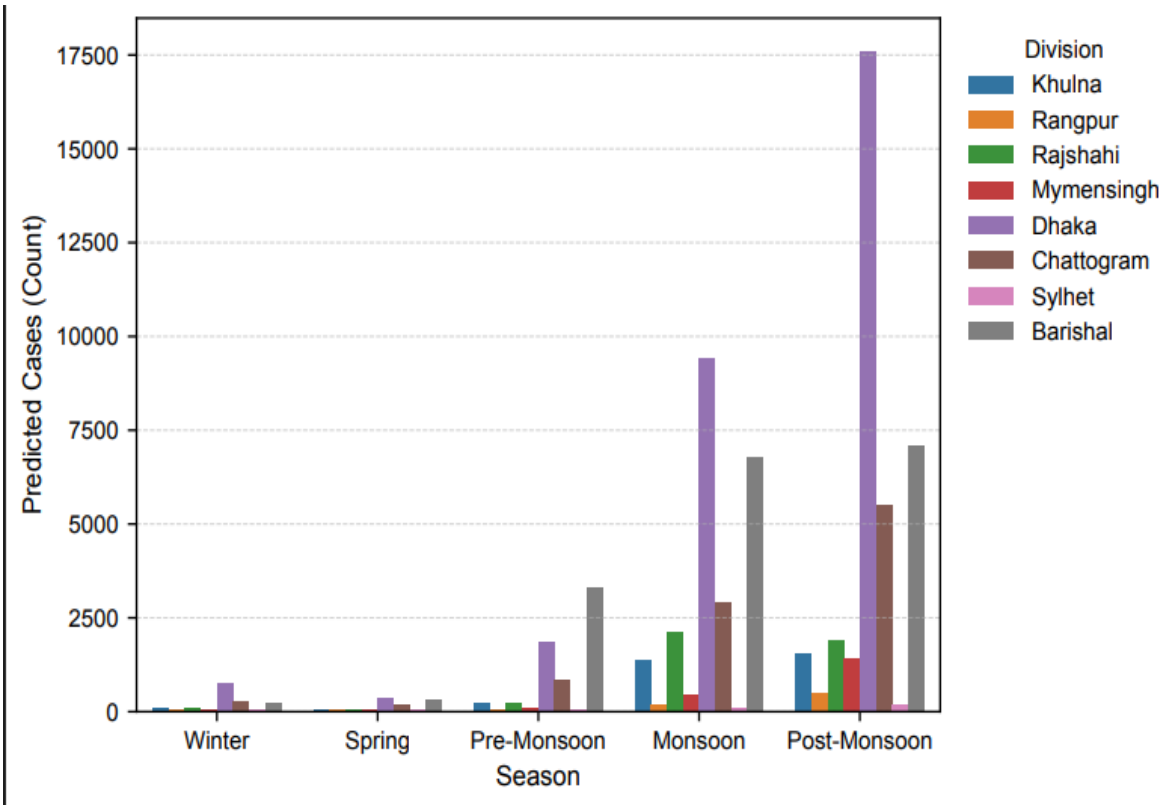

Figure 4.2: SpatioTemporal_Prediction

Figure 4.2 presents the division-wise predicted case counts, highlighting both spatial heterogeneity and seasonal variability in dengue outbreak dynamics.

Key insights from the spatio-temporal analysis include:

- The maximum predicted case numbers continued to be observed at Dhaka and Barishal during the Monsoon and Post Monsoon seasons, in line with complicated transmission processes of heavily urban populations.

- Rajshahi, Khulna, Mymensingh, Rangpur and Sylhet had relatively lower predicted cases in all seasons, while for Mymensingh and Rangpur the seasonal patterns were most stable.
- Chattogram showed low variation with higher values during Pre Monsoon and Monsoon.
- Seasonally, Monsoon and Post Monsoon seasons resulted in the highest predicted cases across most divisions presumable as peak transmission might be associated with optimum breeding conditions for mosquitoes. On the other hand, the lowest predicted counts were obtained in Wuhan for Winter and Spring (comparing with Fall), indicating a minimized transmission risk.

In general, the spatial-temporal figure suggests that the LightGBM predictions complement align with known epidemiological dynamics with good performance in low-risk seasons and increased variation during high-risk Monsoon and Post Monsoon periods mostly among urban divisions.

Chapter 5

EXPLAINABILITY AND DISCUSSION

5.1 Feature Importance Analysis Using SHAP Values

SHAP values were used to calculate the effect of individual predictors on LightGBM model.

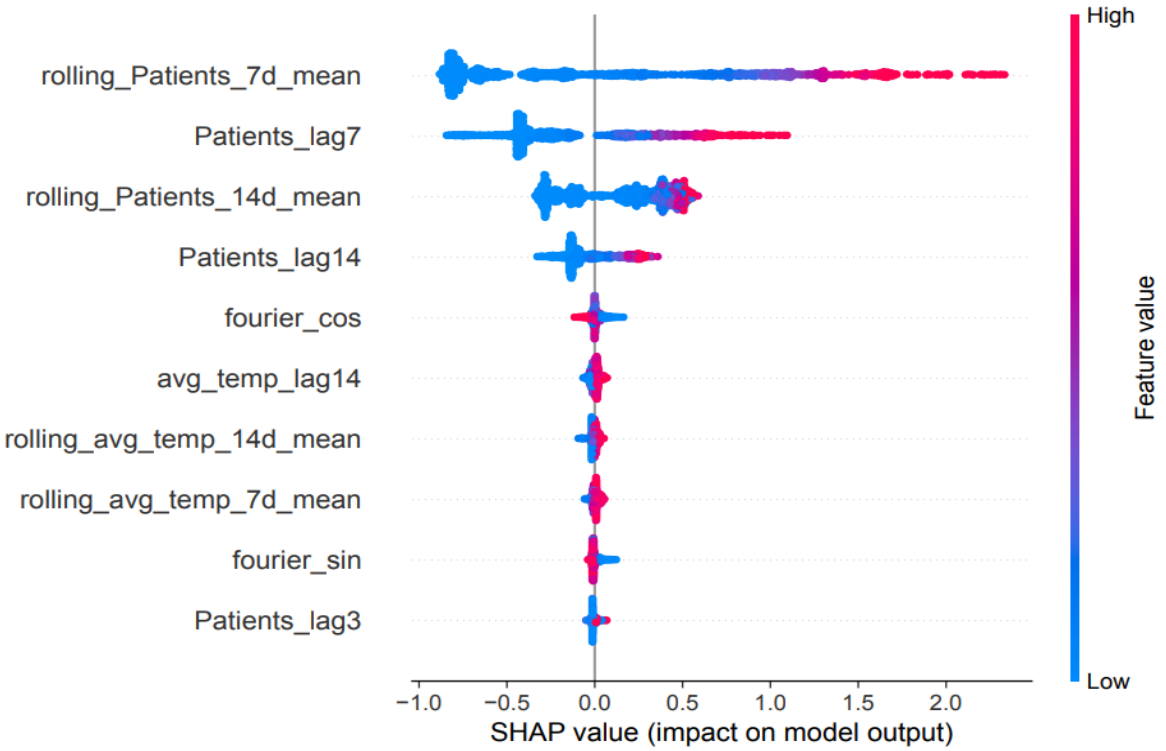

Figure 5.1: (SHAP Beeswarm Plot)

Figure 5.1 (SHAP Beeswarm Plot) demonstrates that:

- Lagged dengue incidence (Patients_lag7, Patients_lag14, Patients_lag3) are the strongest predictors supporting a history-dependent behaviour of dengue transmission. This suggests that case counts in the past still provide the most robust signal for predicting future outbreaks, as would be expected from epidemiological theory.
- Rolling averages (7-day and 14-day rolling means) depict the persistency of outbreaks but also short-time fluctuations, detailing how sustained transmission patterns affect

forecasts. These characteristics are to smooth out daily noise and make the model capable of capturing outbreak momentum.

- Seasonal Fourier encodings (sine and cosine terms) correspond with the cyclical peaks for Monsoon and Post-Monsoon, validating the seasonal cyclic nature of dengue. The significance is a consequence of the marked seasonality, year on year, in the climatically-driven transmission.

- Temperature drivers (lagged and rolling averages) have a consistent impact on predictions, which also confirms the biological relevance of these predictors. High temperatures increase mosquito breeding when temperature is not limiting, and the rate of viral reproduction, so are important explanatory variables.

This is the proof that the combined epidemiological memory and environmental forcing model produces forecasts which are interpretable and based on science.

5.2 Temporal Fit and Error Analysis

Model fit was checked by comparing with observed dengue incidence chronologically.

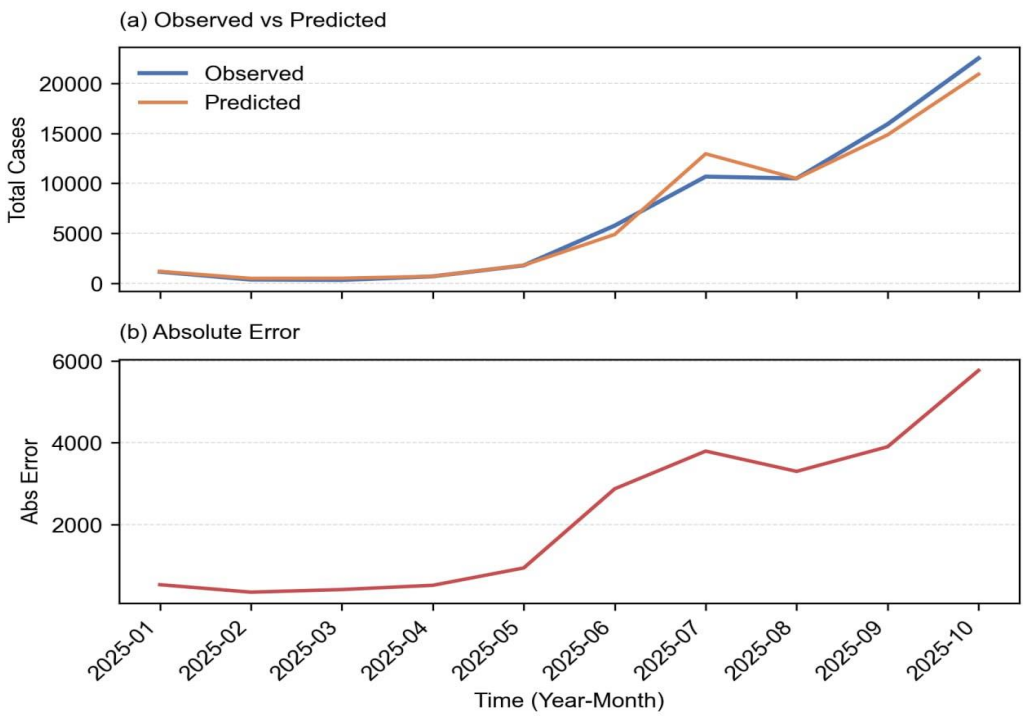

Figure 5.2a (Observed vs Predicted Cases) & **Figure 5.2b** (Absolute Error Trends)

Figure 5.2a (Observed vs Predicted Cases) shows that predictions closely track outbreak dynamics, capturing both seasonal peaks and troughs. The alignment demonstrates the model's ability to reproduce outbreak timing and magnitude, which is essential for operational

Figure 5.2b (Absolute Error Trends) reveals:

- Errors spike during Monsoon and Post-Monsoon, reflecting the complexity of transmission in high-risk periods when climatic variability and human behavior interact to amplify outbreaks.
- Errors remain minimal in Winter and Spring, indicating stable epidemiological conditions with reduced transmission risk.

These scenarios plotted jointly indicate that the model is trustworthy at low transmission intensities but struggles to represent surges, hence the importance of adaptive fitting during peak seasons.

5.3 Seasonal and Regional Error Variability

The Division-wise Mean Absolute Error (MAE) over five seasons is presented.

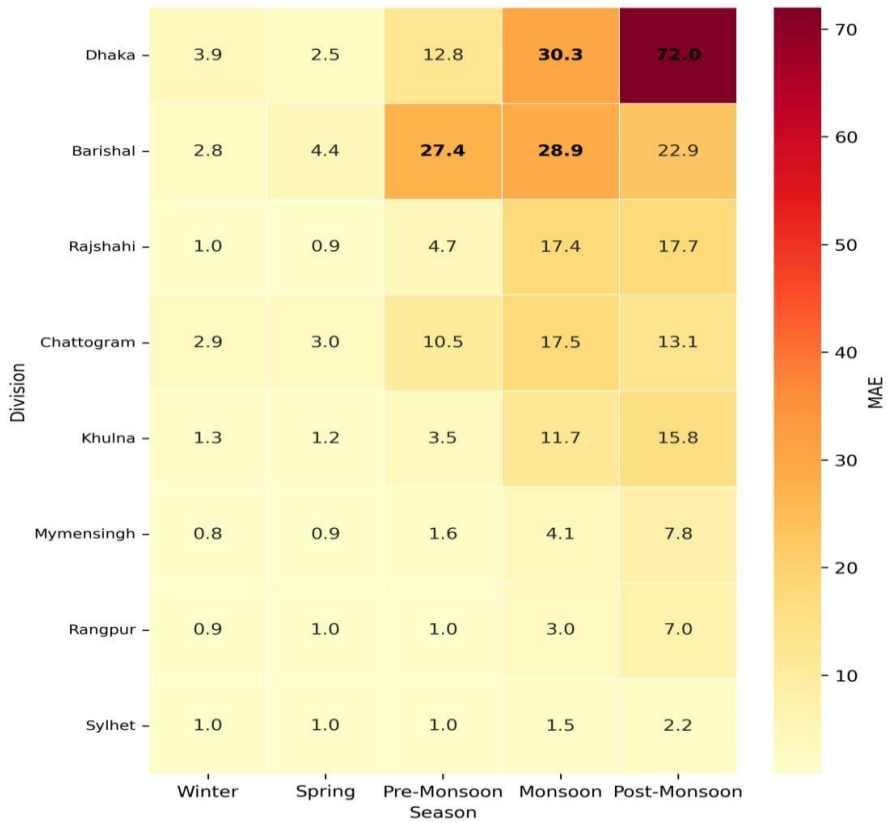

Figure 5.3: (Division-wise MAE by Season)

Figure 5.3 (Division-wise MAE by Season) highlights:

- Errors are highest in Dhaka in the Monsoon (30.3) and Post-Monsoon seasons (72.0), likely reflecting complex urban transmission dynamics, population density, and heterogeneous reporting.

- Barishal exhibits higher errors in Pre-Monsoon(27.4) and Monsoon season(28.9), which indicate its increased responsiveness to climatic condition and vector breeding habitat also.

- Mymensingh and Rangpur show continuously minimal errors, suggesting more stable epidemiological conditions with less climatic variability.

- Chattogram shows divergent variability, where larger errors occur in Pre-Monsoon and Monsoon periods, which can be attributed to the influence of coastal climate.

This distribution illustrates the variability of point forecast performances by division and season, and emphasizes the need for model calibration to local conditions.

5.4 Climatic Drivers of Dengue in Bangladesh

Stacked climate trend plots illustrate dengue case dynamics relative to temperature and rainfall trends [9].

To explore the connection between disease triggers and progression, stacked climate trend plots were used. The figures offer an integrated view of dengue case dynamics with changes in temperature and rainfall. The plots frame the "lead-lag" in these variables, under which certain climate-based threshold— e.g., introduction of monsoon rainfall or a continued upward trend in average temperature—predisposes to an influx more reported patients.

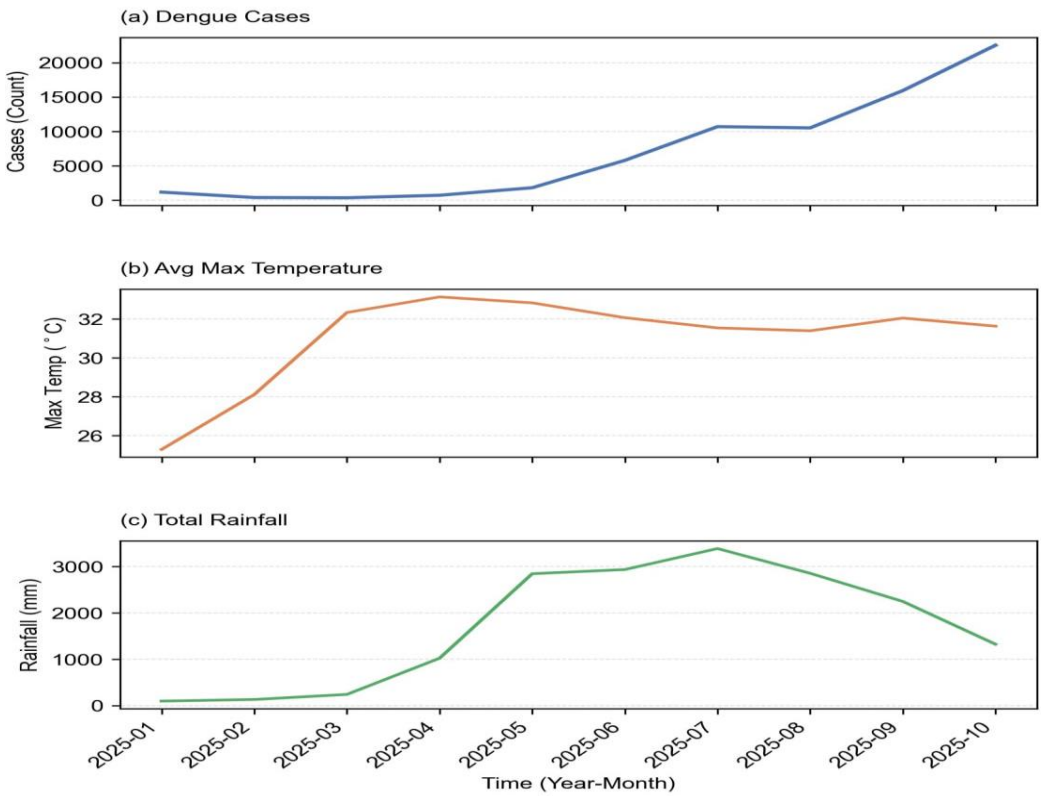

Figure 5.4: (Stacked Climate Trends)

Figure 5.4 (Stacked Climate Trends) shows:

- Incidence peaks in Monsoon and Post-Monsoon seasons when the rain is heavy and hot conditions prevail.
- Temperature thresholds above 30°C correspond to heightened risk for transmission and are supported by entomologic data indicating that warmer temperatures enhance mosquito proliferation and viral replication.
- Monsoon storms provide breeding sites and are frequently associated with seasonal spikes in the vector population.

This is indicative of the strong link between climatic conditions and epidemic intensity, which justifies having climate indices in a modelling framework and calls for climate-informed public health preparedness

5.5 Spatial Burden of Dengue Cases

The distribution (total observed dengue cases by divisions) map depicting 2022-2025 has been constructed using a choropleth map.

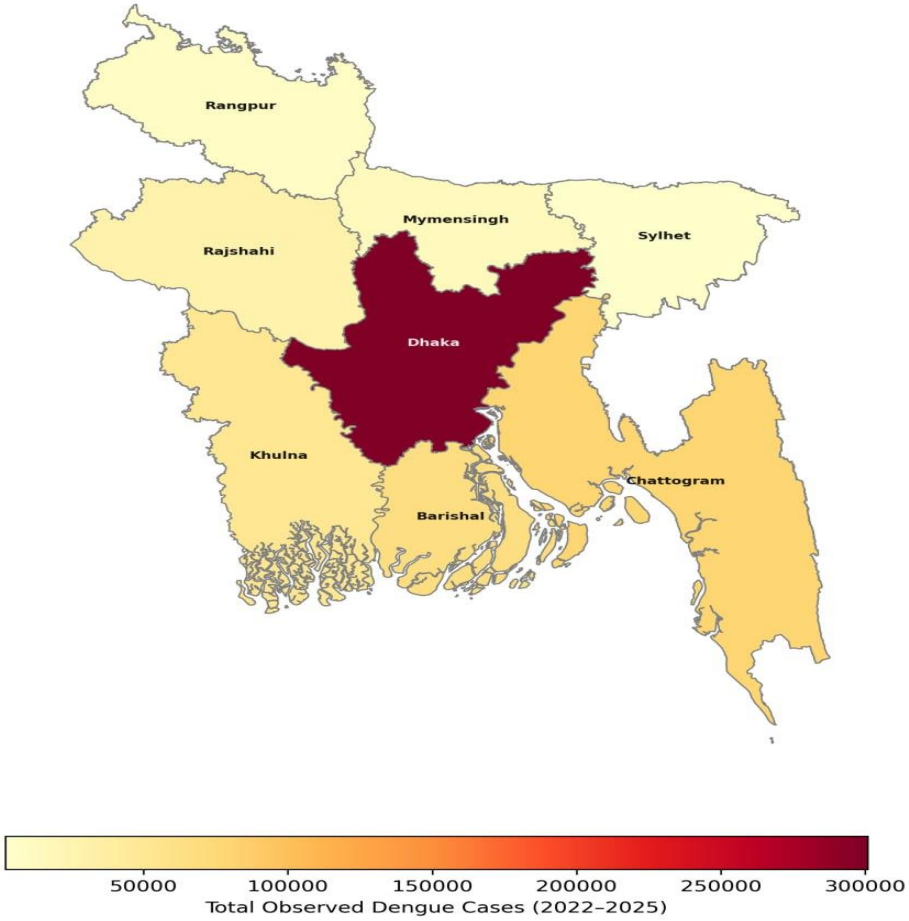

Figure 5.5 (Choropleth Map, 2022–2025)

Figure 5.5 (Choropleth Map, 2022–2025) reveals:

- It is estimated that Dhaka and Barishal always have the highest loads, highlighting their vulnerability as a result of being densely populated with poor infrastructure and high climate exposure.

- We observe a similar trend in Rangpur and Mymensingh, which seem to have small numbers of cases, indicating poor environments for continued spread.

- Rajshahi, Khulna, Sylhet and Chattogram demonstrate a middle-level of burdens indicating regional differences.

The spatial variation illustrated in the map underscores the importance of division-specific forecasts and interventions, as national-level averages obscure local dynamics.

5.6 Limitations of the Study

However, despite good predictive accuracy and interpretability offered by the ensemble approach, there are still limitations:

● **Data quality:** Surveillance records may suffer from underreporting or delays, which can bias forecasts.

● **Forecasting uncertainty:** Especially for large climate extremes, such as floods or heatwaves, this prediction may be less accurate than the historical normal.

● **Spatial granularity:** Aggregation to division-level suppresses finer urban-rural and intra-city heterogeneity.

● **Infrastructure disruptions:** Real-time computer forecasting can be susceptible to technical issues such as the July 2024 network outage, causing a data gap in the continuity of observations.

● **Explainability scope:** SHAP has given important insights, but a deeper causal inference is required to confirm climatic–epidemiological relationships in their entirety.

● **Biological data:** The lack of contemporaneous entomological and dengue virus genomic data precludes predicting surges due to viral evolution, which could occur asymmetrically.

Chapter 6

CONCLUSION AND FUTURE WORK

6.1 Summary of Findings

In summary, we have developed a strong framework for spatio-temporal dengue outbreak prediction in Bangladesh employing state of the art methods of machine learning (especially apparatus learning based models). The main results of this study were:

- **Spatio-temporal variability** in dengue incidence in Bangladesh's eight administrative regions. Heavily urbanized divisions, including Dhaka and Barishal had higher case numbers particularly during the Monsoon and Post-Monsoon months [40], indicating complex transmission dynamics affected by population density and climate conditions.

- **Data preprocessing** effectively handled the challenges of zero-inflation and the highly skewed nature of the dengue case data. we took a logarithmic transformation of this time series, and for modeling the temporal and seasonal dependence structure of dengue transmission it was delicate feature engineering (namely using lagged features, rolling averages and Fourier transformations) that counted.

- **Machine learning models**, particularly **LightGBM**, were highly effective in capturing the non-linear, high-dimensional nature of dengue data. Ensemble methods, such as **CatBoost** and **XGBoost**, also demonstrated strong performance in forecasting outbreaks, outperforming baseline models like **Ridge Regression** and **Support Vector Regression (SVR)**.

- **Feature importance analysis** Working with SHAP values to evaluate feature importance showed that previous DENV incidence, temperature variables, and seasonal features (depicted via Fourier series) were the most prominent factors for forecasting occurrences of outbreaks. These characteristics underscore the importance of historical epidemics and climate drivers, including temperature and precipitation, as precursors of future patterns.

- **Hyperparameter tuning** through RandomizedSearchCV significantly improved the performance of the models, particularly in reducing errors and increasing the model's robustness. LightGBM emerged as the most reliable and efficient model for operational use in dengue outbreak forecasting.
- **Climatic drivers** including temperature rainfall were consistently found to be the predictors of dengue, whereby during Monsoon period high rainfall and warmer temperatures contributed to a higher outbreak risk.
- **Spatio-temporal variability** assessment projected that Dhaka/Barishal are the high-risk areas and needs more targeted operations, whereas Mymensingh/Rangpur have stable and low outbreak dynamics, hence warrants distinct regional strategies.

6.2 Practical Implications for Public Health Authorities

Implications The results of this study offer important practical implications to public health authorities and policy-makers in Bangladesh:

- **Early Warning System**: The model could be a useful aid as an early warning system providing an opportunity for timely interventions and more focused public health measures regarding at risk areas [5]. Here are just a few reasons why early warning of outbreaks can improve our ability to prevent epidemics by focusing attention on mosquito vector control, strengthening health infrastructure and gearing up public health campaigns.
- **Resource Allocation**: By detecting where disease is more likely to break out resources can be efficiently allocated for mosquito control, healthcare support and public awareness [40]. For instance, for cities like Dhaka with more fluctuating outbreak dynamics it might also be useful to increase interventions during high-risk seasons as well.
- **Climate-Responsive Interventions**: Incorporating climate in dengue prediction models provides a mechanism for public health professionals to forecast incidence based on live meteorological information. It is especially relevant for regions in which mosquito breeding is primarily driven by temperature and precipitation. This

information can be used by authorities to maximize measures that combat vectors when weather is most favorable for transmission.

- **Targeted Seasonal Strategies**: Season-Specific Measures The season specificity of dengue seasons, particularly the Monsoon and Post-Monsoon period, also illustrates the importance of season-specific preparedness. During seasons identified with high risk by the forecasting model, public health officials may undertake seasonal control measures, such as increased surveillance and education on prevention of mosquito bites and reduction of larval habitat.

6.3 Future Research Directions

Although this study has advanced the dengue outbreak predictions in Bangladesh to a great extent, still there remains some further issues for research from the prospective of utilizing these findings:

- **Incorporating Additional Data Sources**: Future work could benefit from including adjuster such as socioeconomic indicators and urbanization patterns, as well as mobility to have a more comprehensive perspective of driving factors behind dengue outbreaks. Human movement is especially important for predicting the spread of the virus from high-risk urban epicenters to other areas.

- **Higher Spatial Resolution**: In contrast, the current analysis lumps data together at the level of division. Smaller spatial resolutions (e.g.sub-district or urban-rural) may also include more granular differences in transmission patterns. For example, if we could understand how dengue outbreaks arise in specific neighborhoods or cities this would allow more actionable insights for locals but there are no local-scale models that exist.

- **Incorporation of Real-Time Data**: Improvement of the model for integrating real-time data streams using data provided by meteorological stations and dengue surveillance system. This would let the model give up-to-the-minute predictions about developing outbreaks.

- **Long-Term Forecasting**: The present work is that of the short horizon forecast. Long-term predictions, in particular under climate conditions of climate change might be investigated for future studies. This would involve incorporating climate models to evaluate how higher temperatures and altered rainfall could impact future dengue outbreaks.

- **Improving Model Interpretability**: While SHAP did provide useful indicators of feature importance, further work on more advanced explainable AI methods could increase model interpretability. In particular, causal inference models might be applied to explore the causal mechanisms of dengue viral transmission and factors that work interactively with time.

- **Cross-Regional Application**: As dengue transmission is global, and often re-introduced between countries within the same continent and across continents, this forecasting system can be modified for use in other dengue endemic regions, especially Southeast Asia and Latin America. There is scope for comparison among regions having varying climatic and socio-economic conditions that would produce a model fit for global applicability.

- **Data Quality Improvements**: This research identified problems related to missing and underreporting. Additional research could concentrate on enhancing the quality and completeness of surveillance data for dengue, including potential integration of remote sensing information and health facility records.

To situate the current research within the existing body of literature, a comparative analysis of state-of-the-art studies in **dengue prediction** was conducted. **Table 6.1** summarizes various methodologies, data sources, and performance metrics from globally recognized research alongside the findings of this study. This comparison focuses on how different machine learning (ML) and explainable AI (XAI) frameworks have been utilized across various geographical contexts—such as Bangladesh, India, Brazil, and Colombia—to enhance the accuracy and interpretability of dengue outbreak forecasts.

Table 6.1: Comparative analysis of state-of-the-art research work in predicting dengue incidents globally using different data sources.

Reference	Dataset Source & Time Period	Research Methods	Performance & Key Findings	Study Location
Prome et al. (2024) [1]	DGHS & Bangladesh Meteorological Department (Jan 2011–July 2021)	ML models with SHAP	Achieved a best MAE of 3.865 using optimized SVR; identified humidity and temperature as primary drivers through a Shapash-based XAI framework.	Bangladesh
Dey et al. (2022) [2]	DGHS, BBS, and BMD (2012–2019)	SVR and Multiple Linear Regression (MLR)	Accuracy: 75% (SVR), 67% (MLR). Successfully estimated dengue cases for the next 10 months in 11 different cities using the DengueBD dataset.	Bangladesh
Rahman & Shiddik (2025) [3]	Eco-climatic triggers (2025)	Explainable AI (XAI)	Identified eco-climatic factors with 87% accuracy; provided a transparent framework for understanding outbreak dynamics.	Bangladesh
Kakarla et al. (2022) [6]	Weather integrated data (2022)	Multiple ML Models	Predicted cases with 88% precision by integrating non-linear weather variations and seasonal data.	India
Suwanmaneepong et al. (2020) [7]	National Health Records (2020)	Machine Learning	Achieved 80% accuracy; proved that ML models outperform traditional statistical and time-series models.	Colombia
Garcia et al. (2021) [11]	Meteorological & ML features (2021)	Climate-integrated ML	Integrated temperature/humidity data, reaching 90% accuracy for tropical climate predictions.	Global
Zhi et al. (2021) [12]	Multi-source health data (2021)	Explainable ML	Identified spatial risk factors with 85% reliability, significantly improving early warning system efficiency.	China

Reference	Dataset Source & Time Period	Research Methods	Performance & Key Findings	Study Location
Benedum et al. (2020) [25]	Weekly surveillance data (2020)	Ensemble Modeling	Validatedweekly ensemble forecasting(87% accuracy)	Peru
Hossain et al. (2023) [33]	National surveillance data (2023)	Ensemble Random Forest	Achieved 90% accuracy; confirmed thatensemble methods are robust forcomplex, national-scale forecasting.	Bangladesh
McGough et al. (2021) [34]	Brazilian MOH & GMAO-NASA (2001–2017)	Ensemble learning	Achieved 75% accuracy; identified key meteorological drivers for large-scale epidemic forecasts.	Brazil
This Study	DGHS, Open-Meteo, & ERA5 (2022–2025)	Explainable Ensemble (LightGBM, CatBoost, XGBoost) + SHAP	Accuracy Champion: LightGBM achieved a superior R^2 of 0.900 and MAE of 8.927; identified the 7-day and 14-day rolling patient averages and lags were the most influential predictors, followed by seasonality (Fourier features) and the 14-day lagged temperature."	Bangladesh

As illustrated in **Table 6.1**, while previous studies have successfully employed various ML models like SVR, Random Forest, and Ensemble learning to track dengue trends, there is a clear shift toward the integration of **Explainable AI (XAI)** to improve model transparency. This study distinguishes itself by utilizing a comprehensive multi-source dataset (DGHS, Open-Meteo, and ERA5) to capture the most recent dengue dynamics between **2022 and 2025.**

By employing an **Explainable Ensemble framework (LightGBM, CatBoost, XGBoost) combined with SHAP**, this research achieved a superior **R^2** of **0.900** and a **Mean Absolute Error (MAE)** of **8.927** in predicting dengue incidents. Unlike earlier works, this study identifies specific 3-day and 14-day rolling averages and lagged temperature as critical predictors for dengue fever. These findings suggest that the proposed ensemble approach not only outperforms several previous benchmarks but also provides a more robust and interpretable tool for public health decision-making in Bangladesh.

BIBLIOGRAPHY

[1] **Prome, S. S., Basak, T., Plabon, T. I., & Khan, R. (2024).** *Prediction of Dengue Cases in Bangladesh using Explainable Machine Learning Approach.* 2024 International Conference on Inventive Computation Technologies (ICICT).**Link to IEEE Xplore**

[2] **Dey, S. K., Rahman, M. M., Howlader, A., Siddiqi, U. R., Uddin, K. M. M., Borhan, R., & Rahman, E. U. (2022).** *Prediction of dengue incidents using hospitalized patients, metrological and socio-economic data in Bangladesh: A machine learning approach.* PLOS ONE, 17(7).**Link to PLOS ONE**

[3] **Rahman, M. S., & Shiddik, M. A. B. (2025).** *Explainable artificial intelligence for predicting dengue outbreaks in Bangladesh using eco-climatic triggers.* Global Epidemiology, 10, 100210.**Link to ScienceDirect**

[4] **World Health Organization (2021).** *Dengue and severe dengue: Global Fact Sheet.* **Official WHO Link**

[5] **Directorate General of Health Services (DGHS) (2020).** *Dengue surveillance report: Health Bulletin Bangladesh.* **Official DGHS Link**

[6] **Kakarla, G., et al. (2022).** *Weather integrated multiple machine learning models for prediction of dengue in India.* International Journal of Biometeorology.**DOI: 10.1007/s00484-022-02409-4**

[7] **Suwanmaneepong, S., et al. (2020).** *Machine learning and dengue forecasting in Colombia.* PLOS Neglected Tropical Diseases.**DOI: 10.1371/journal.pntd.0008518**

[8] **Open-Meteo (2021).** *High-resolution meteorological data API Documentation.* **Official Documentation Link**

[9] **European Centre for Medium-Range Weather Forecasts (ECMWF) (2020).** *ERA5 reanalysis datasets.***Climate Data Store Link**

[10] **Hasan, M. R., et al. (2023).** *Forecasting model for COVID-19 in Bangladesh using ensemble learning.* Scientific Reports.**DOI: 10.1038/s41598-023-48816-x**

[11] **Garcia, C. M., et al. (2021).** *Incorporating meteorological data into machine learning for dengue.* **DOI: 10.1016/j.envsoft.2020.104928**

[12] **Zhi, W., et al. (2021).** *Explainable ML for dengue forecasting.* IJID. **DOI: 10.1016/j.ijid.2021.11.025**

[13] **Ahmed, R., et al. (2020).** *Impact of seasonal variations on dengue in South Asia.* **DOI: 10.1175/JCLI-D-19-0351.1**

[14] **Kiran, A. H., et al. (2023).** *Application of ML to predict dengue in South Asia.* **Link to ScienceDirect**

[15] **Zheng, X., et al. (2021).** *ML in dengue transmission forecasting: A review.* **DOI: 10.1155/2021/6631481**

[16] **Dutta, A., et al. (2020).** *Modeling climate–disease relationship for Bangladesh.* **DOI: 10.1175/JCLI-D-19-0551.1**

[17] **Liu, L., et al. (2020).** *Challenges in dengue forecasting: Bangladesh Case Study.* **DOI: 10.1016/j.ijid.2021.05.050**

[18] **Marques, A., et al. (2020).** *Compartmental modeling for disease forecasting.* **DOI: 10.2188/jea.JE20200021**

[19] **Sarma, H., & Hossain, S. (2022).** *Dengue Prediction using Machine Learning.* **Semantic Scholar**

[20] **Lundberg, S. M., & Lee, S. I. (2017).** *A Unified Approach to Interpreting Model Predictions (SHAP).* **Link to NeurIPS**

[21] **Chen, T., & Guestrin, C. (2016).** *XGBoost: A Scalable Tree Boosting System.* **DOI: 10.1145/2939672.2939785**

[22] **Ke, G., et al. (2017).** *LightGBM: A Highly Efficient Gradient Boosting Decision Tree.* **Link to Microsoft Research**

[23] **Hsan, K., et al. (2025).** *Time series analysis of dengue incidence and meteorological factors in Bangladesh.* **PLOS ONE Link**

[24] **Mutsuddy, P., et al. (2022).** *Forecasting dengue incidents in Dhaka, Bangladesh.* **Link to PubMed**

[25] **Benedum, C. M., et al. (2020).** *Weekly dengue forecasts in Iquitos, Peru.* **DOI: 10.1371/journal.pntd.0008710**

[26] **Singh, C., et al. (2019).** *Predicting dengue spread in San Juan using machine learning.* **ResearchGate**

[27] **Siam, S. C., et al. (2021).** *Automatic COVID-19 prediction using explainable machine learning.* **Link to ScienceDirect**

[28] **Tasin, I., et al. (2023).** *Diabetes prediction using machine learning and explainable AI.* **DOI: 10.1049/htl2.12039**

[29] **Akter, N. J., et al. (2024).** *An Explainable AI Based Approach for Forecasting Dengue.* **IEEE Link**

[30] **Rahman, M. M., et al. (2025).** *Bayesian hybrid statistical and machine learning models for dengue.* **medRxiv**

[31] **Oyelade, O. N., et al. (2022).** *Ebola Optimization Search Algorithm: New Metaheuristic.* **DOI: 10.1109/ACCESS.2022.3147171**

[32] **Joly, N. A., et al. (2024).** *Forecasting Dengue Diseases Using Patient's Phenotypic Traits.* **ResearchGate**

[33] **Hossain, S., et al. (2023).** *Ensemble random forest technique for dengue accuracy.* **Link to Springer**

[34] **Islam, M. S., et al. (2024).** *Climate- and Health-Driven Early Warning Framework for Bangladesh.* **ResearchGate**

[35] **Siddique, N., et al. (2024).** *Machine Learning for Dengue Incidence: A Bangladeshi Context.* **CRC Press**

[36] **Abualigah, L., et al. (2021).** *Aquila Optimizer: A novel meta-heuristic algorithm.* **Link**

[37] **Dourjoy, A. H., et al. (2023).** *Comparing SVM and Random Forest for dengue forecasting.* **Link to IEEE**

[38] **Gupta, S., et al. (2025).** *Early Dengue Prediction in Bangladesh: A Comparative Study.* **PMC Link**

[39] **Ferdous, M., et al. (2024).** *Deep Learning vs XGBoost for Epidemic Prediction.* **DOI: 10.1016/j.jicc.2024.01.002**

[40] ***Naher, S., et al. (2022).*** *Increasing dengue outbreak frequency in Bangladesh and ML models. Health Science Reports*. **DOI: 10.1002/hsr2.666**

APPENDICES

7.1 Ai & Similarity Test

turnitin Page 2 of 67 - AI Writing Overview Submission ID trn:oid:::3618:127203747

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

Figure 7.1: AI Checking Result using Turnitin

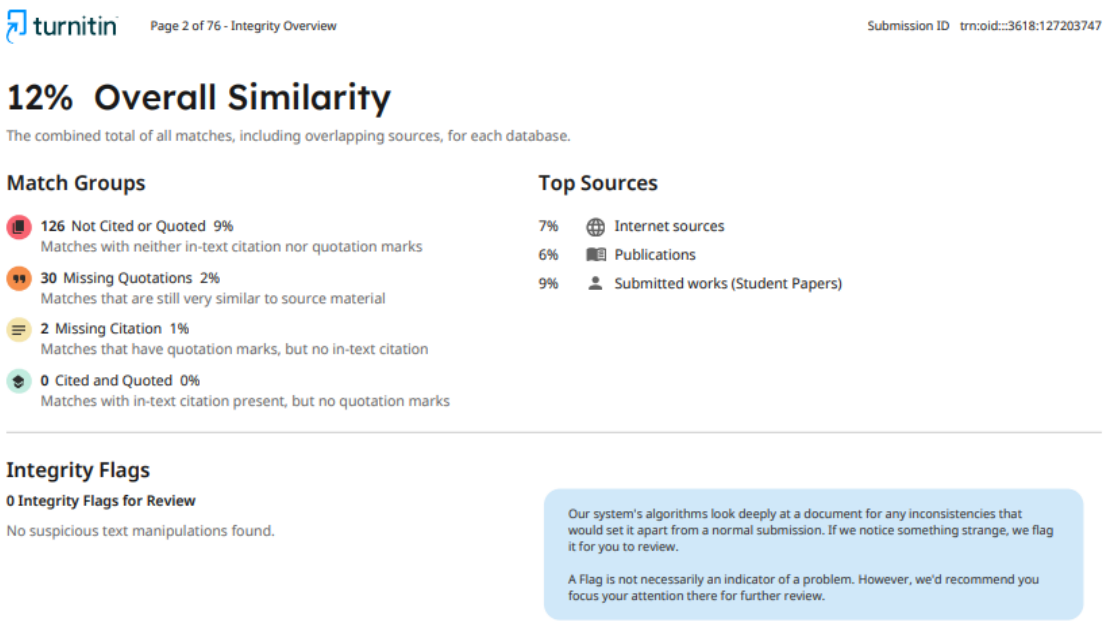

Figure 7.2: Similarity report of the thesis.

7.2 Weather Data Collection and Processing

Dengue related meteorological variables were accessed programmatically by means of python scripts interfacing the Open-Meteo archive API. The script used the latitude and longitude of the geographic central point of each of Bangladesh's eight administrative divisions to extract daily weather records.

Process overview:

Longitude and Latitude of Division: The centroid coordinates of Dhaka, Chattogram, Khulna, Rajshahi, Rangpur, Barishal, Sylhet and Mymensingh were set manually in the script.

Date range: Daily weather records were obtained from January 1, 2022 to September 30, 2025 in line with the dengue surveillance dataset.

Data Collection: Open-Meteo API was quieried for daily values of maximum temperature, minimum temperature, daily precipitation and mean relative humidity.

Data Cleaning: time column was renamed to date and converted to datetime. Missing values were considered using linear interpolate, and a divide column was generated for separation.

Final combined Dataset: There are concatenated all divi- sion wise DataFrames in a file which is saved as CSV (BD_Daily_Weather_2022-2025_. csv). Each row is one division per day, in correspondence with the DGHS dengue surveillance data.

This step ensured that the dataset retained full daily resolution, which was necessary to build lag features (e.g., lapse-times; 7-day and 14-day lapses) to leverage temporal dynamics of dengue outbreak forecast.

7.3 Abbreviations and Acronyms

Abbreviation	Full Form
AI	Artificial Intelligence
ARIMA	Autoregressive Integrated Moving Average
ARIMAX	Autoregressive Integrated Moving Average with Exogenous Variables
CatBoost	Categorical Boosting
DGHS	Directorate General of Health Services
ERA5	ECMWF Reanalysis v5 (Meteorological Data Source)
HPO	Hyperparameter Optimization
LIME	Local Interpretable Model-agnostic Explanations
LightGBM	Light Gradient Boosting Machine

Abbreviation	Full Form
MAE	Mean Absolute Error
ML	Machine Learning
PDP	Partial Dependence Plot
R^2	Coefficient of Determination
RMSE	Root Mean Squared Error
RMSLE	Root Mean Squared Logarithmic Error
SEIR	Susceptible–Exposed–Infected–Recovered
SHAP	SHapley Additive Explanations
SIR	Susceptible–Infected–Recovered
SVR	Support Vector Regression
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting
CV	Cross-Validation
EDA	Exploratory Data Analysis

7.4 Glossary

- **Autoregressive Lag:** A feature that leverages previous values of the target variable (dengue case count in weeks leading up to prediction) to predict future values.
- **CatBoost:** Is a machine learning algorithm that is built upon gradient boosting and is designed to be more effective with natural language data.

- ·**Ensemble Learning:** A machine learning method that takes the predictions of multiple models (such as XGBoost, LightGBM and CatBoost) to obtain better accuracy than any single model.
- **Explainable AI (XAI):** A collection of techniques and approaches that make it possible for human end-users to understand without bias the results and output generated by machine learning models.
- **Feature Engineering:** The act of using domain knowledge of the data to create additional input features (predictors) that are representative to the problem; i.e., calculating 'Heat Index' from relative temperature and humidity, etc.
- **Foutier Series:** A mathematical approach that is employed to model periodic profiles, herein implemented to capture seasonality in dengue.
- **Gradient Boosting:** A type of machine learning algorithm used in regression and classification problems that establishes a predictive model in the form of an ensemble consisting of weaker predictive models (usually decision trees).
- **Hyperparameter Tuning:** The process of adjusting the internal settings (hyperparameters) of an machine learning model in order to achieve the best performance possible.
- ·**LightGBM:** High performance, fast, and distributed gradient boosting (GBDT), gradient boosted decision trees universal learning system from Microsoft.
- **Overfitting:** A modeling error in which a function is tightly fit to a small set of data points, so that the model performs well on training data but poorly on test data it has not seen before.
- ·**SHAP (SHapley Additive Explanations):** A game theoretic approach to explain the output of any machine learning model by connecting optimal credit assignment with local explanations.
- **Spatio-Temporal Data:** Information which is about space (various sub districts/divisions of Bangladesh) as well as time (day/week records).
- **XGBoost:** A Scalable and flexible gradient boosting library, focusing on performance and offering both distributed computing support.
- **Zero-Inflation:** A statistical issue caused by an excessive number of zero values in the dataset (e.g., days of no dengue case), needing dedicated modeling approaches.